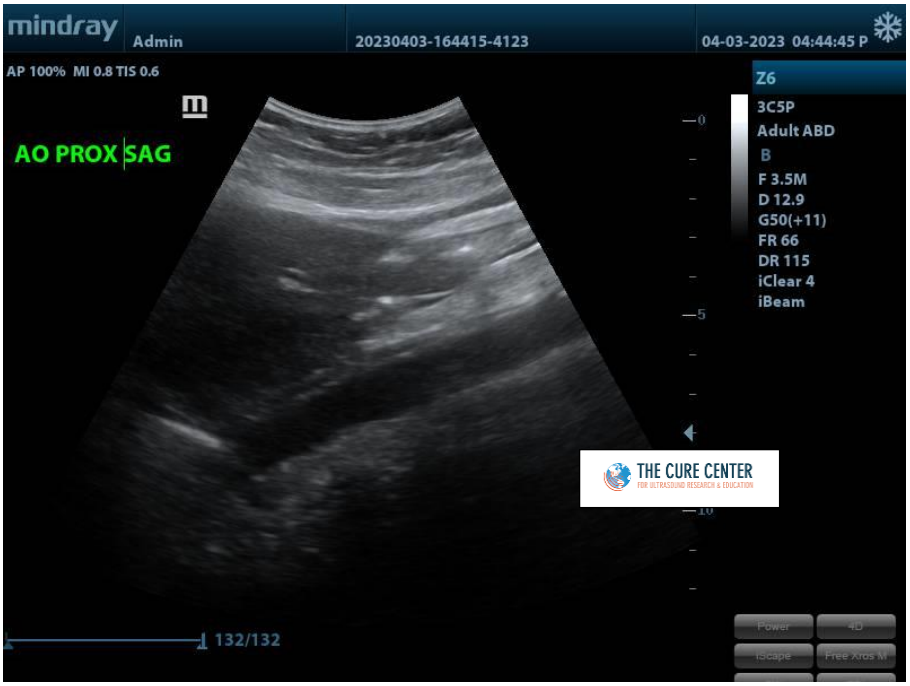
THE CURE CENTER
FOR ULTRASOUND RESEARCH & EDUCATION

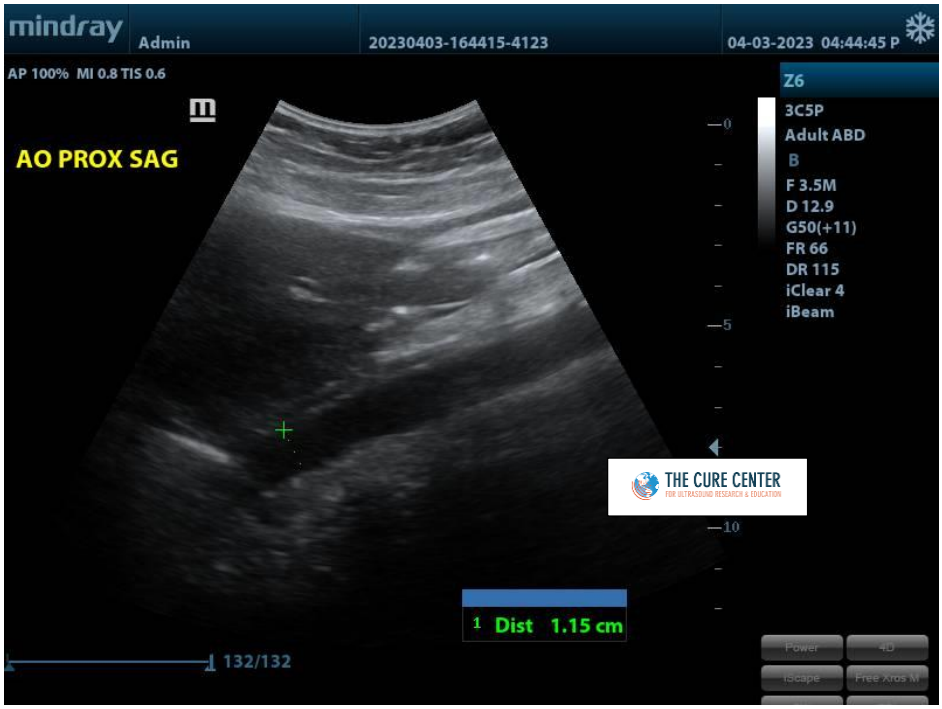
General Abdominal Ultrasound Protocol

Sagittal AORTA

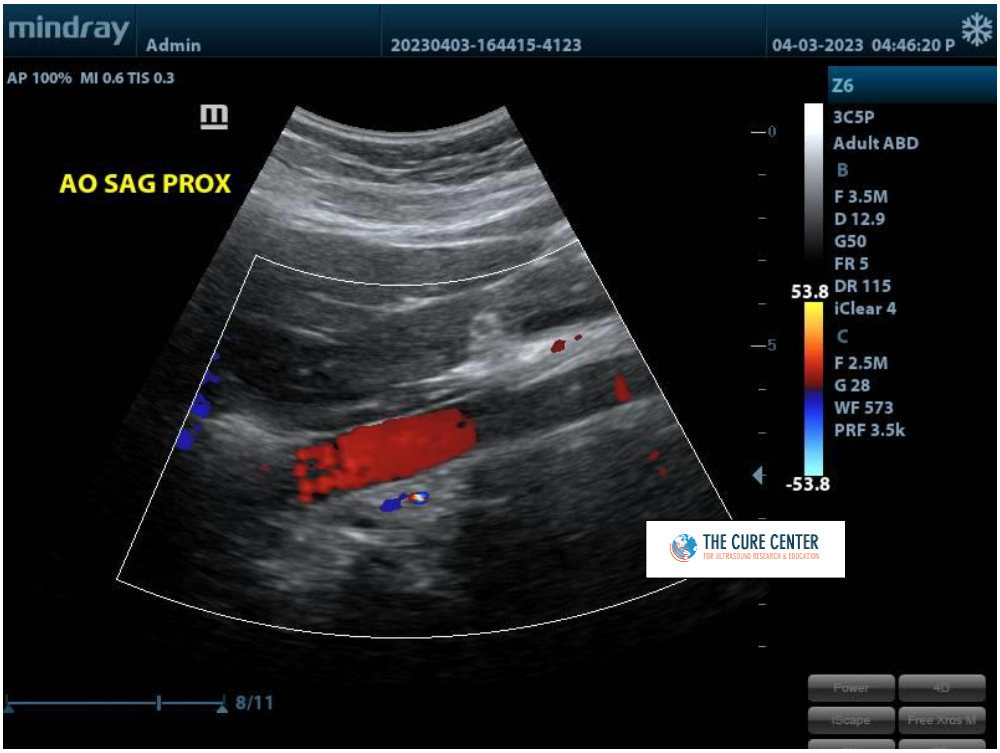
1. SAG AO: Proximal Grayscale



2. SAG AO: Proximal w/ Measurement



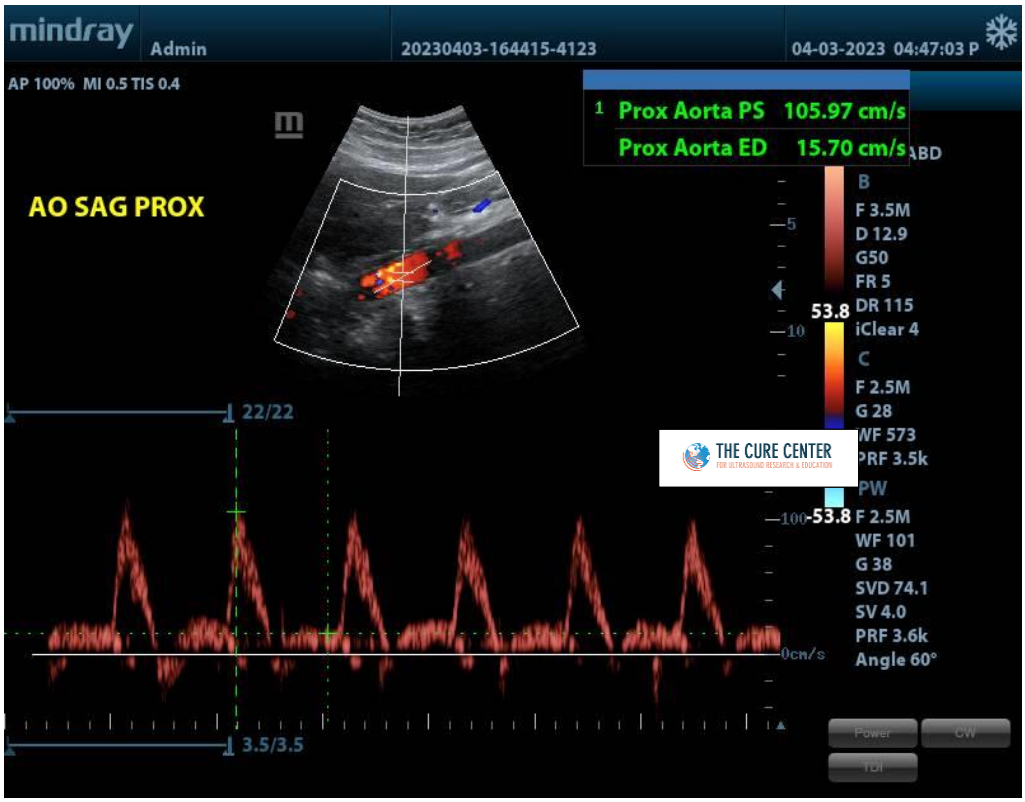
3. SAG AO: Proximal w/ Color



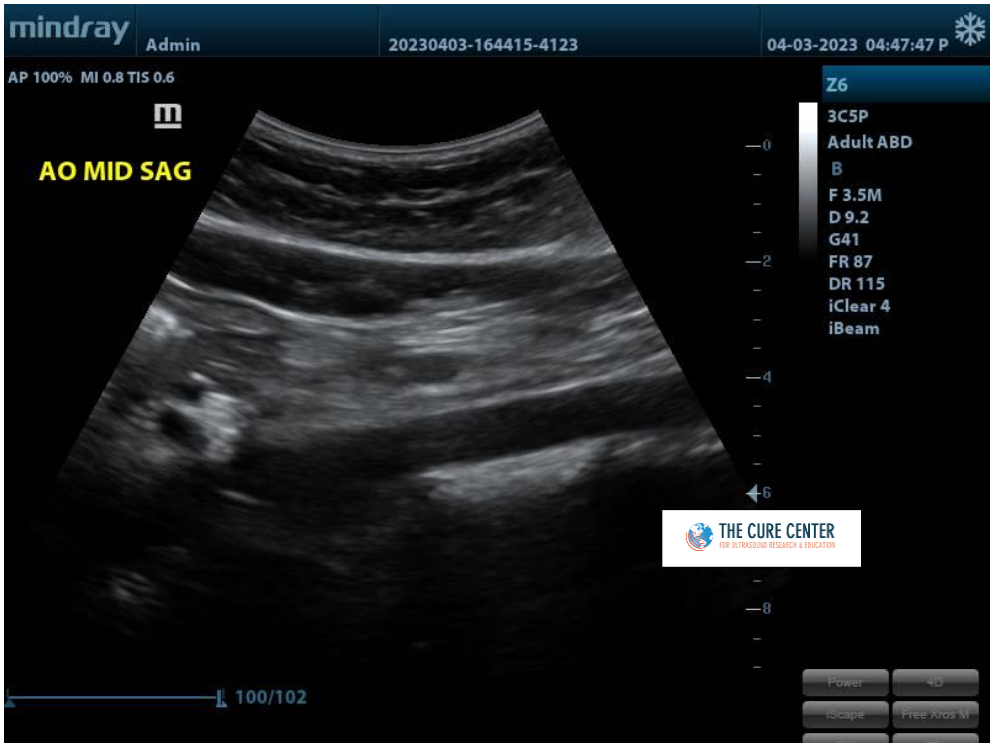
4. SAG AO: Proximal w/ Doppler PSV/ EDV Measurement

NOTE: Be sure to **angle** your sample gate to **<u>60 degrees</u>** for spectral Doppler images.

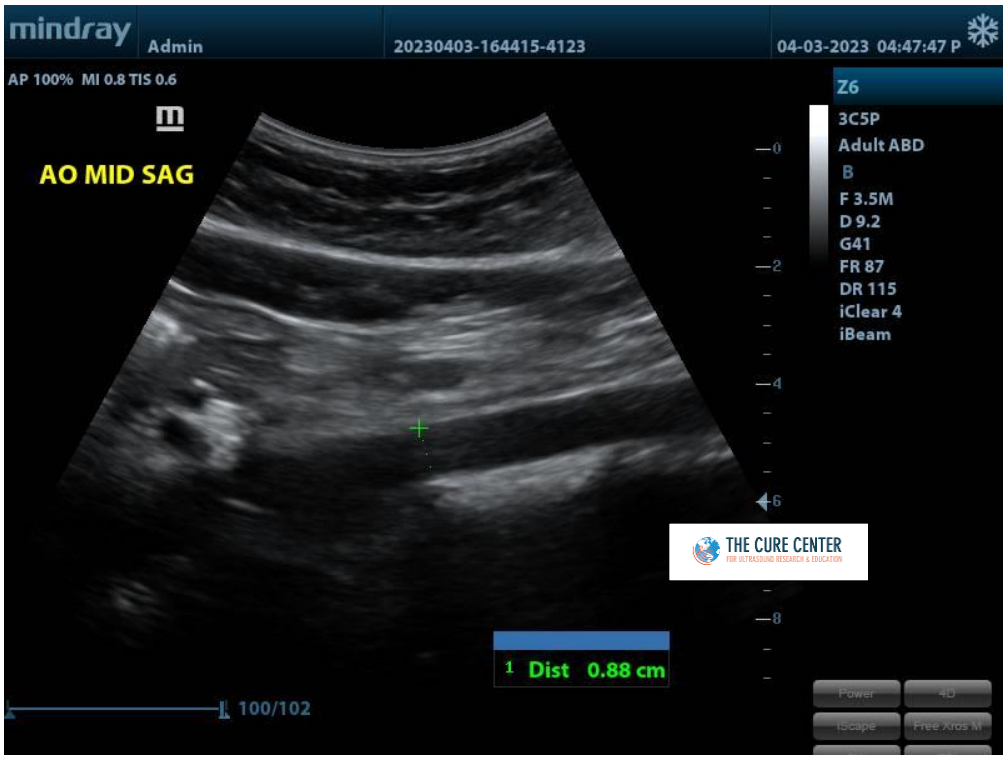
Remember to obtain measurements of the **PSV** and E**DV**, respectively.



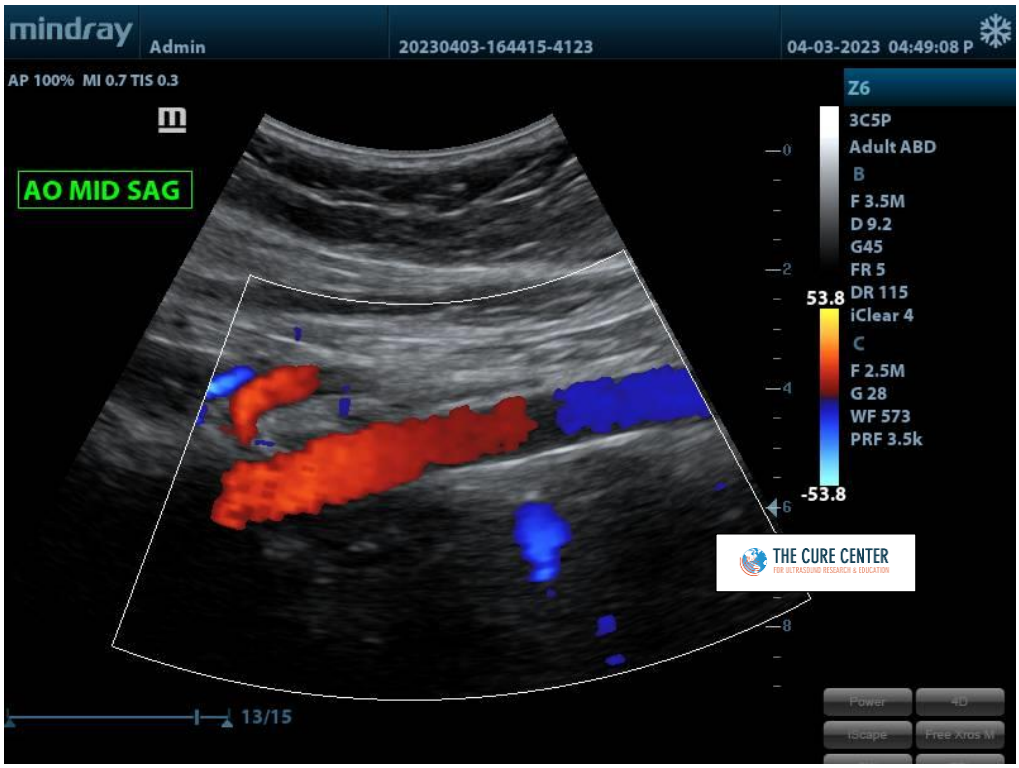
5. SAG AO: Mid Grayscale



6. SAG AO: Mid Grayscale w/ Measurement



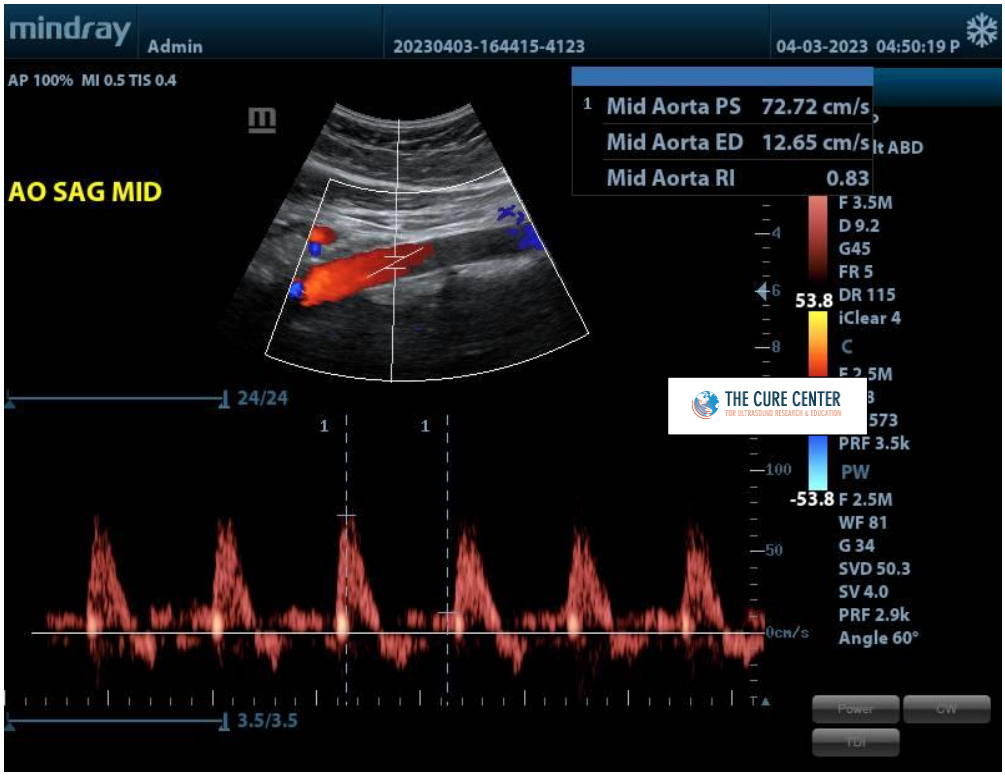
7. SAG AO: Mid w/ Color



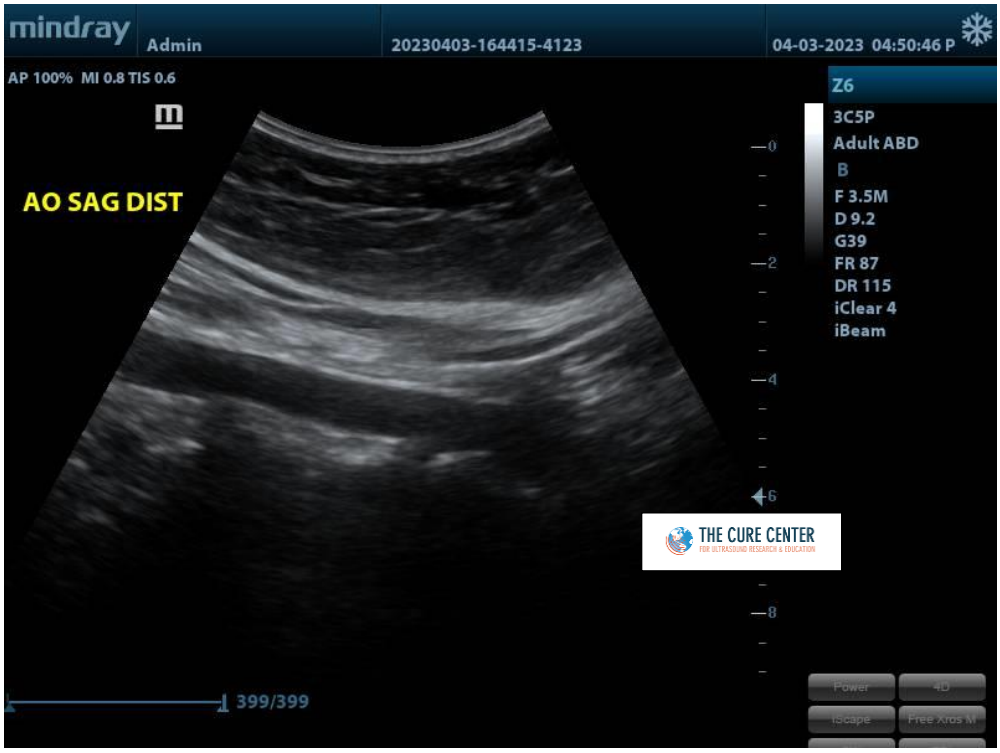
8. SAG AO: Mid w/ Doppler PSV/ EDV Measurement

NOTE: Be sure to **angle** your sample gate to **<u>60 degrees</u>** for spectral Doppler images.

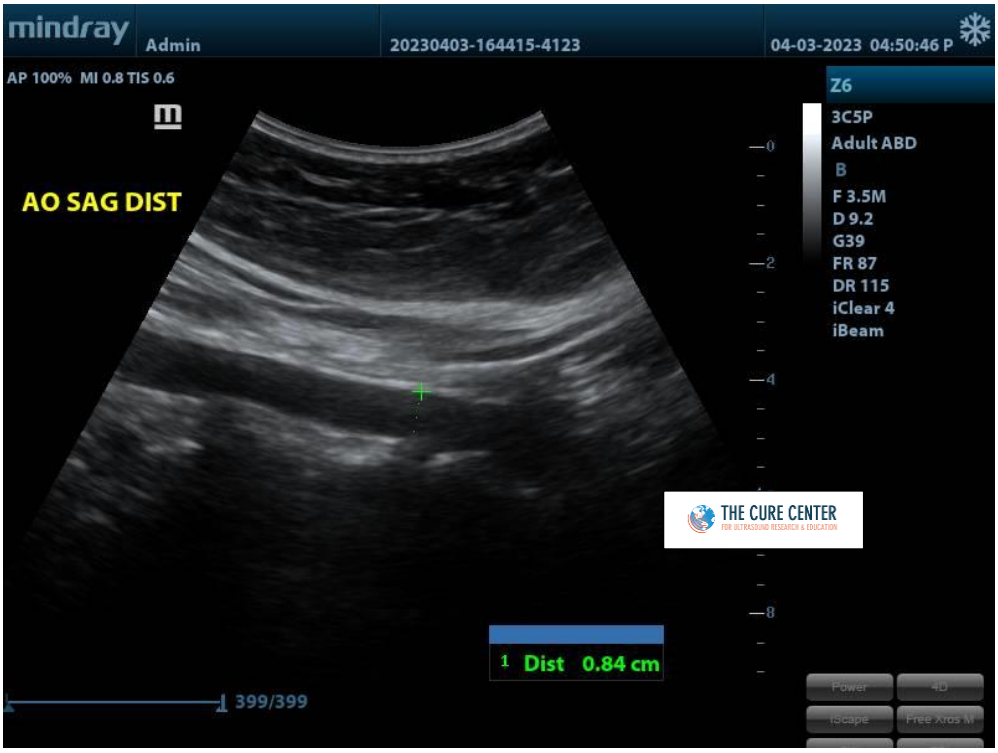
Remember to obtain measurements of the **PSV** and E**DV**, respectively.



9. SAG AO: Distal Grayscale



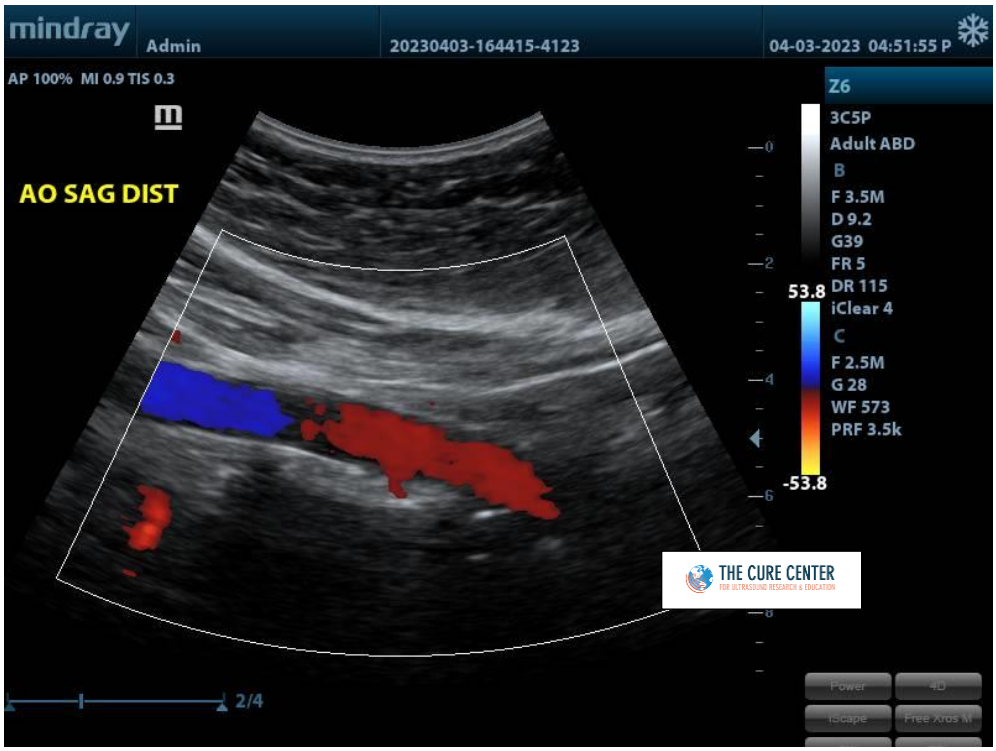
10. SAG AO: Grayscale w/ Measurement



11. SAG AO: Distal w/ Color

NOTE: Remember to invert your color scale from this point on.

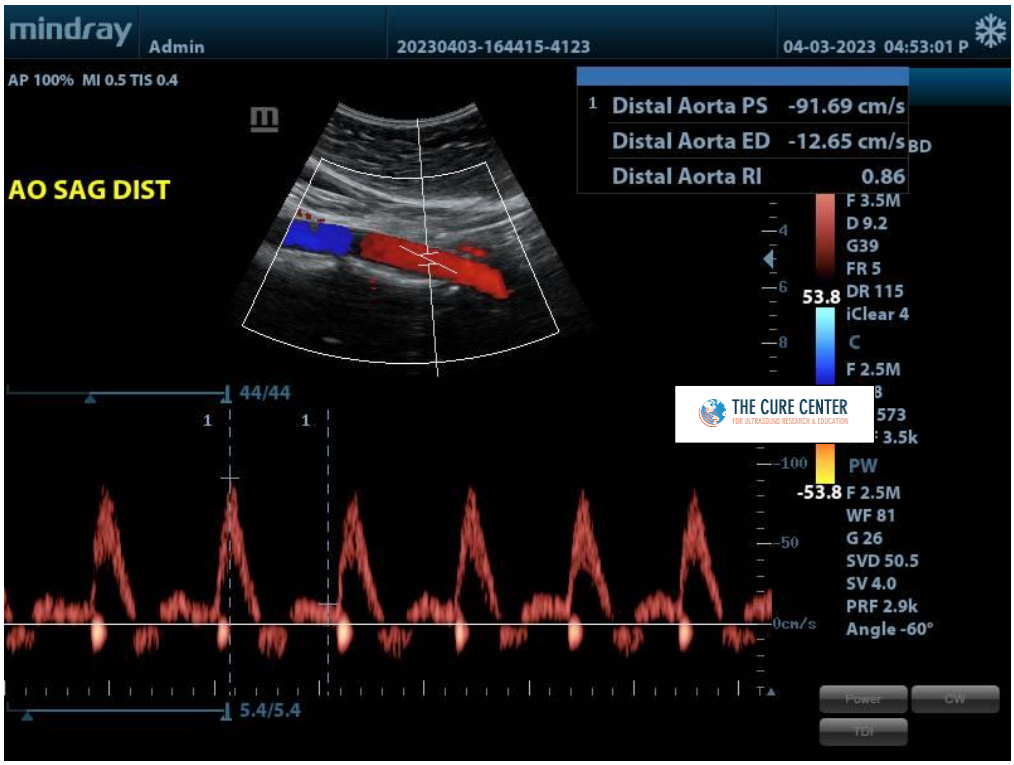
Blue should be on top and red at the bottom.



12. SAG AO: Distal w/ Doppler PSV/ EDV Measurement

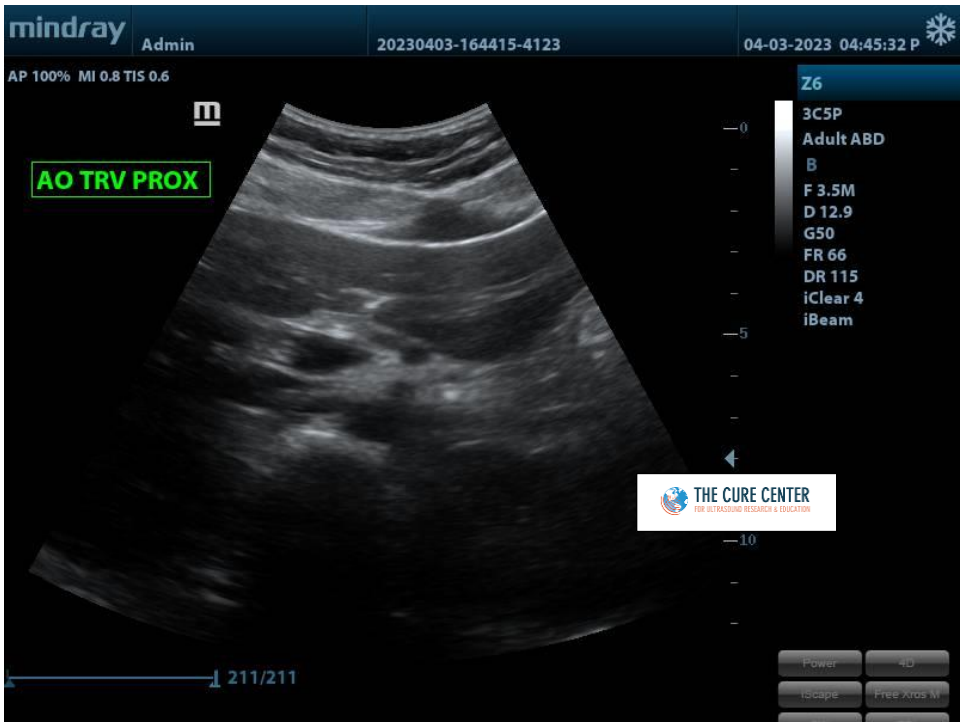
NOTE: Be sure to **angle** your sample gate to **60 degrees** for spectral Doppler images.

Remember to obtain measurements of the **PSV** and **EDV**, respectively.

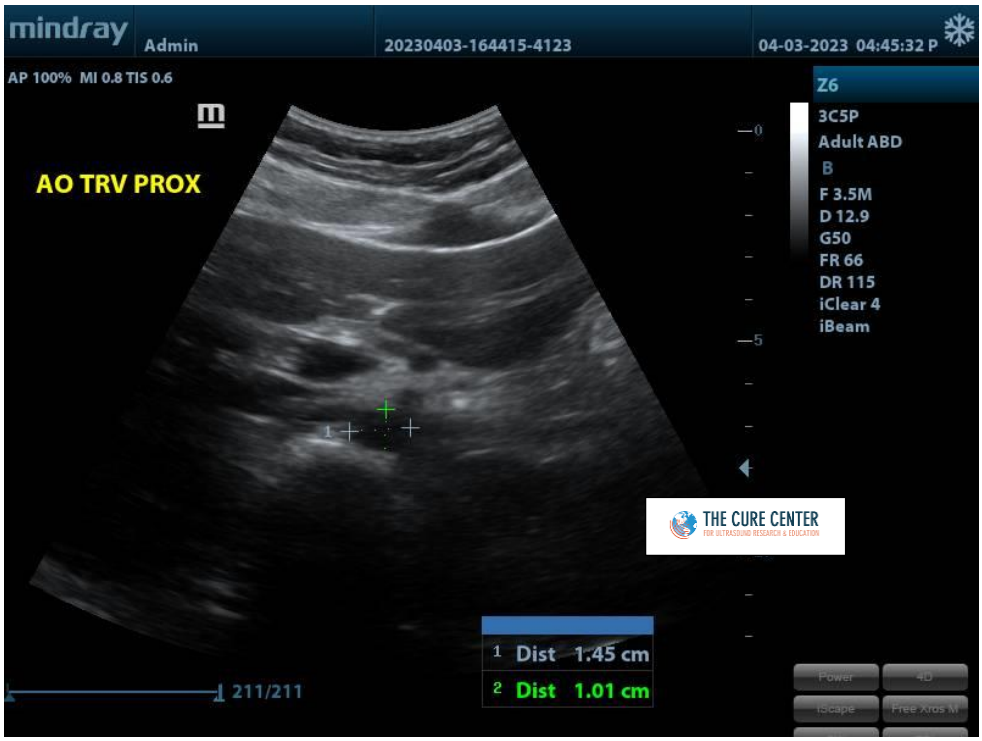


Transverse AORTA

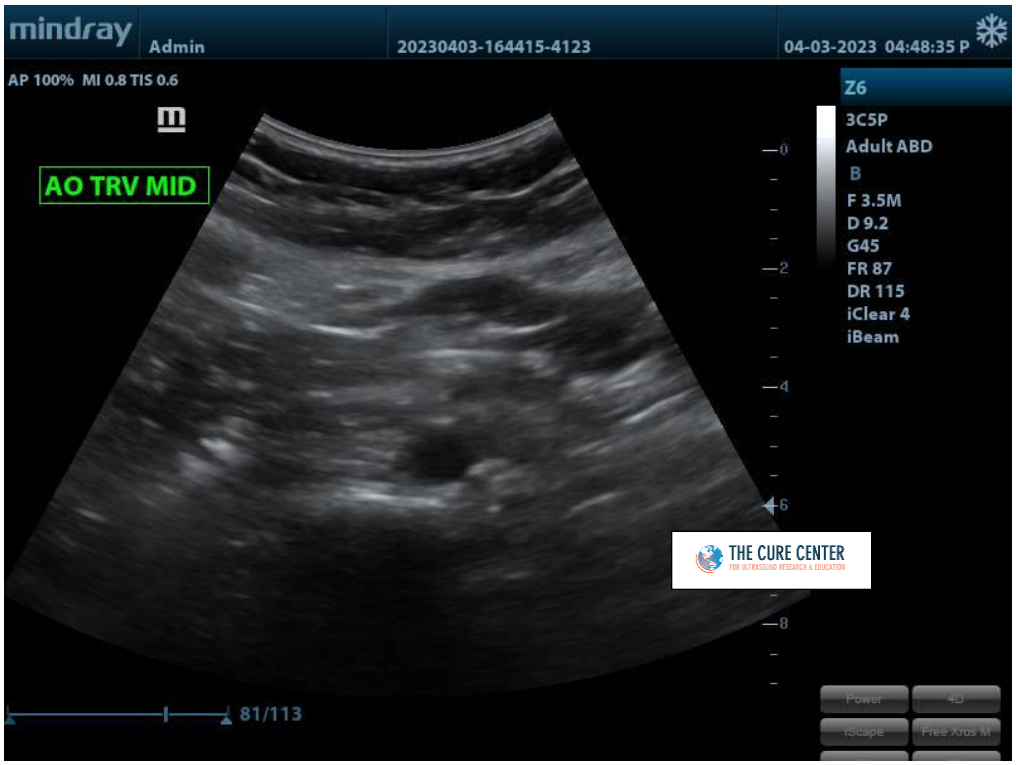
13. TRV AO: Proximal Grayscale



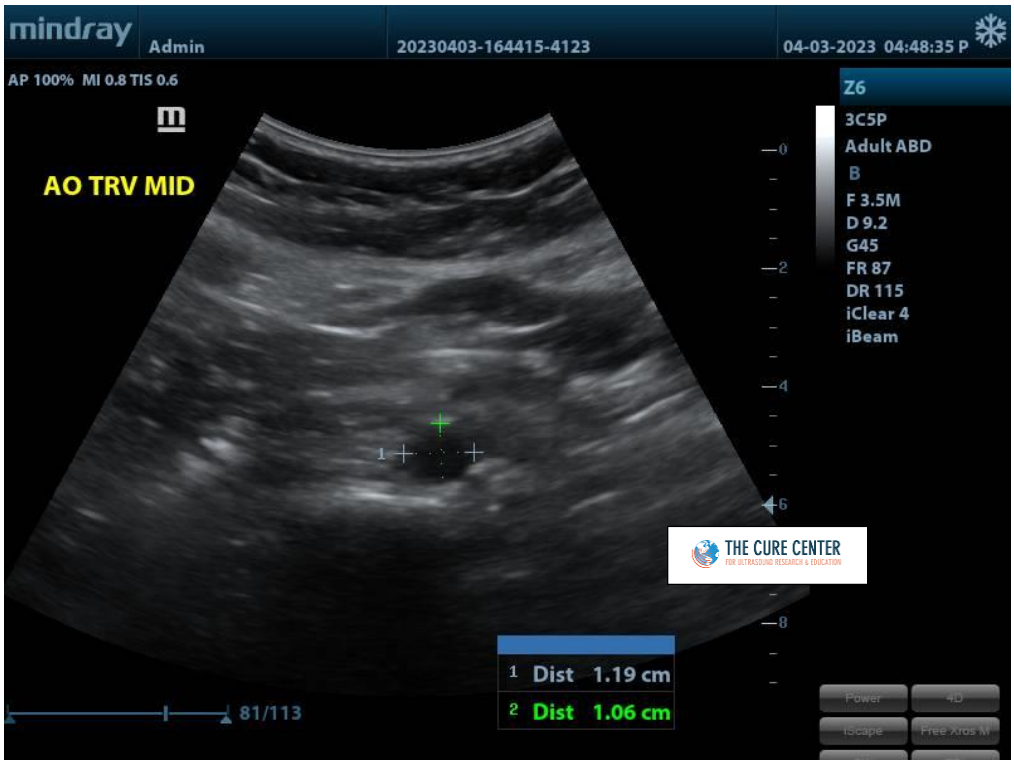
14. TRV AO: Proximal w/ Measurement



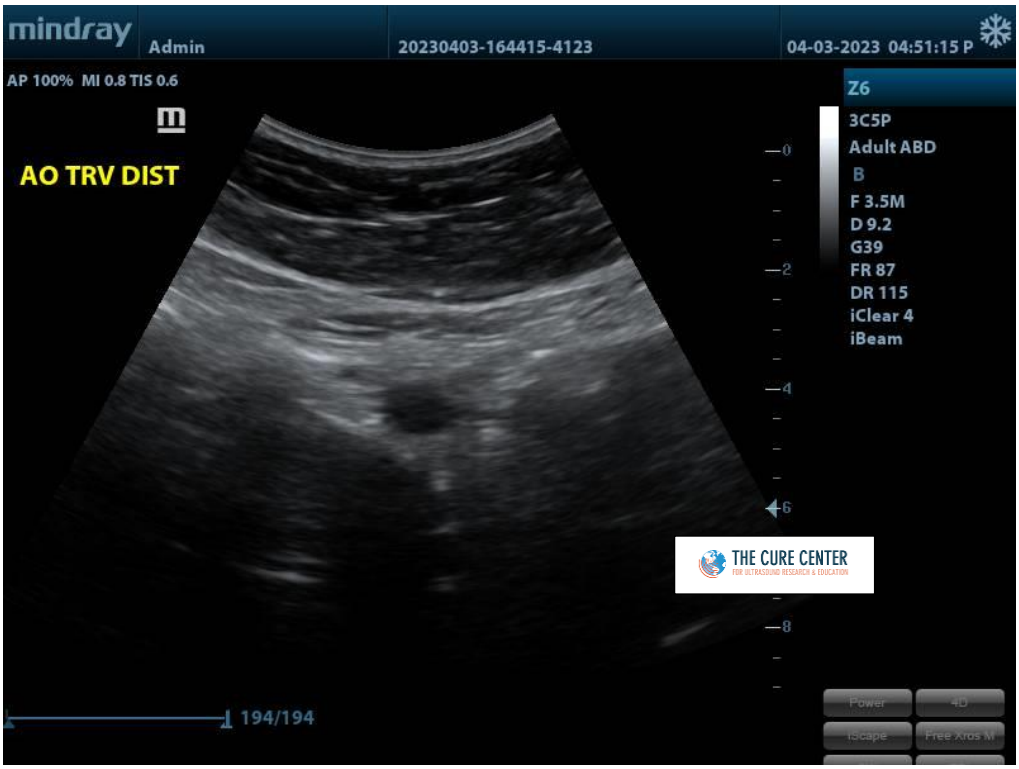
15. TRV AO: Mid Grayscale



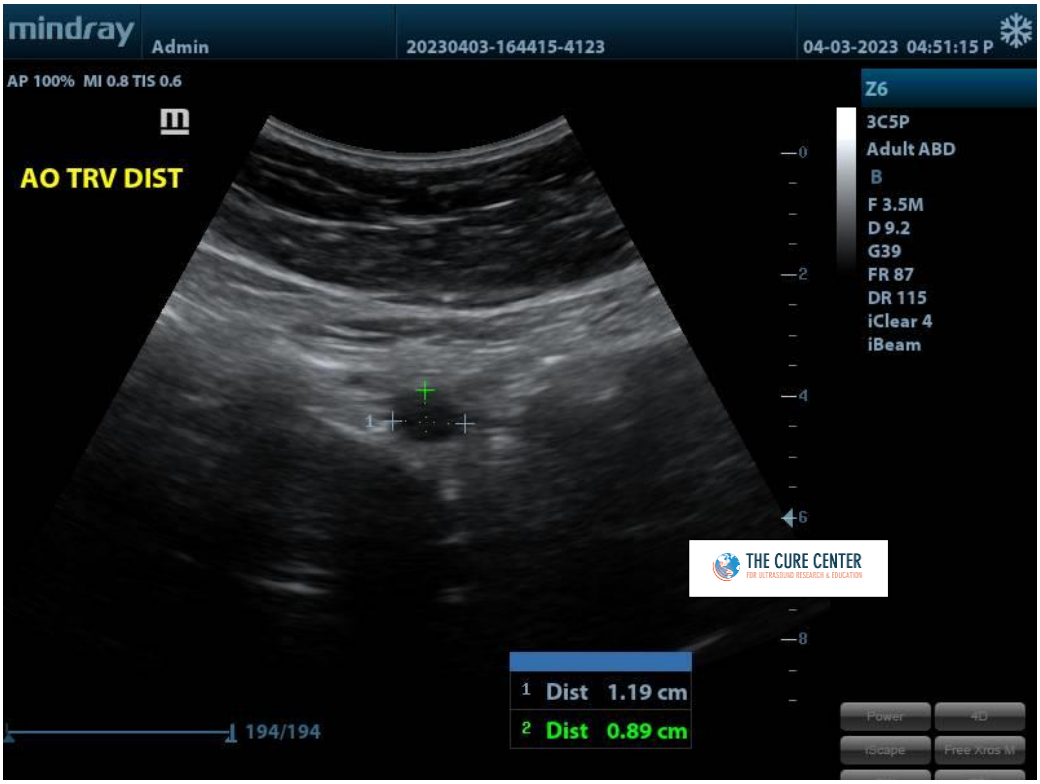
16. TRV AO: Mid w/ Measurement



17. TRV AO: Distal Grayscale

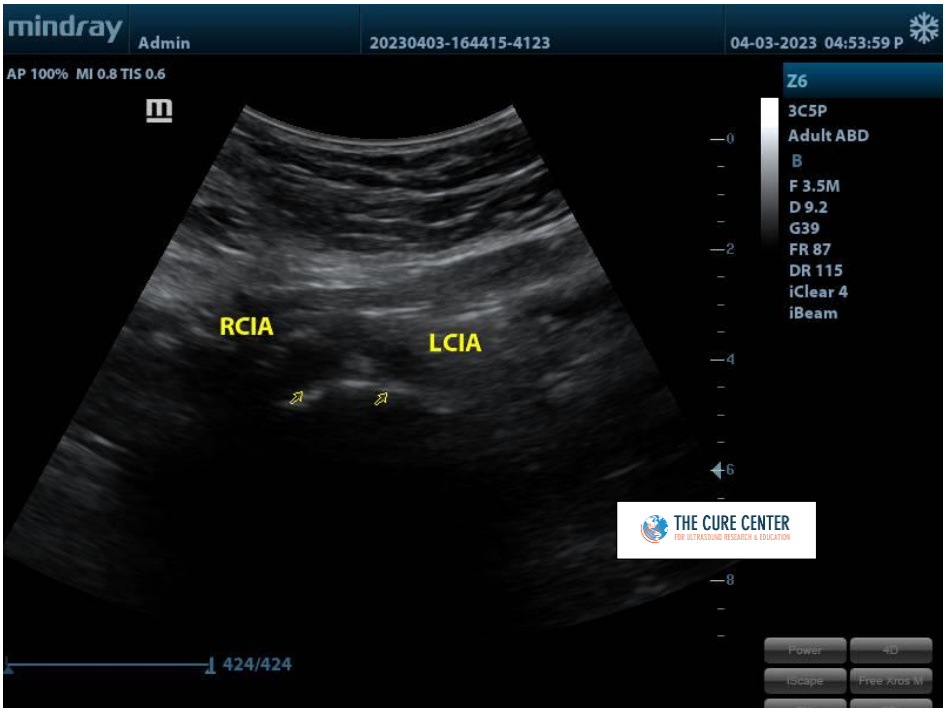


18. TRV AO: Distal w/ Measurement



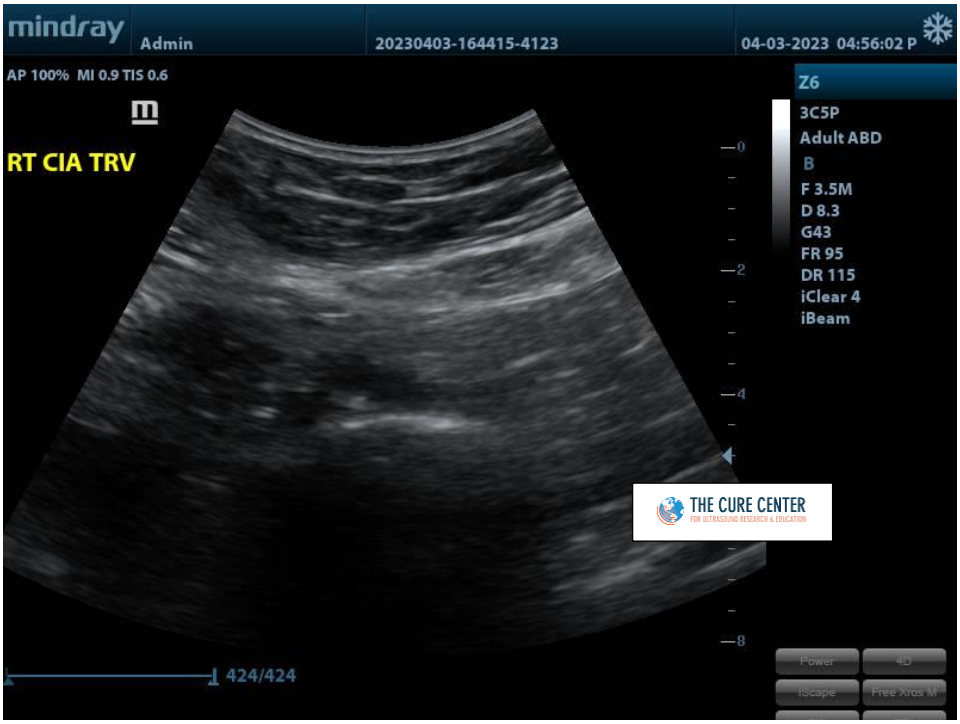
Right + Left COMMON ILIAC ARTERIES

19. TRV RCIA/ LCIA: Grayscale

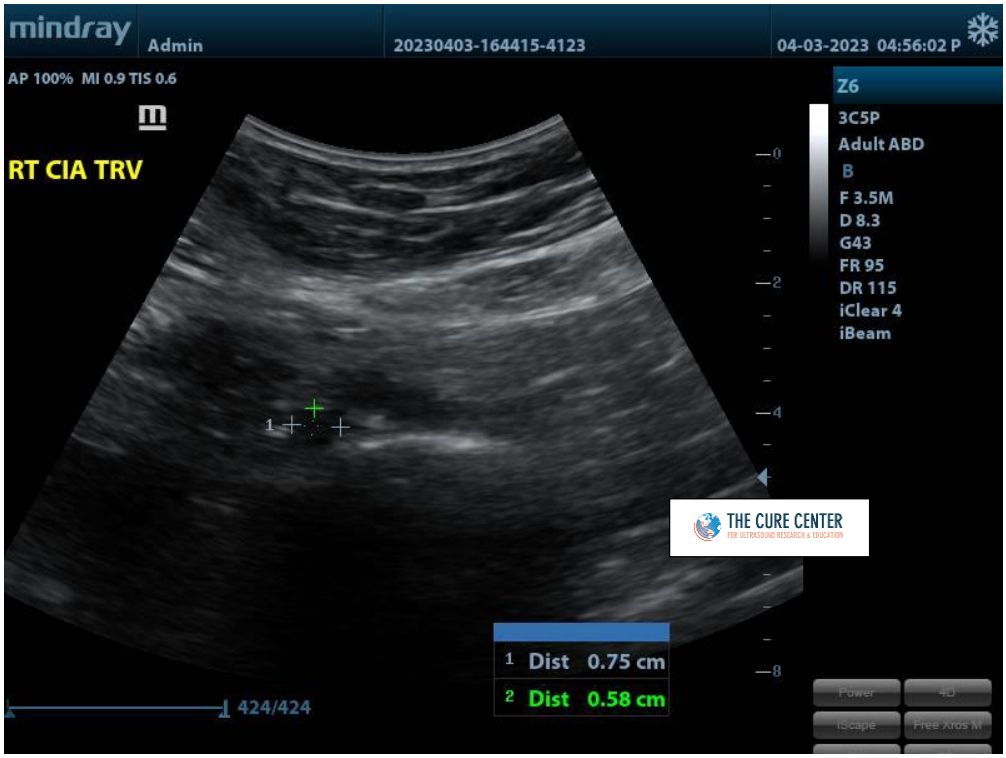


Transverse RT CIA

20. TRV RT CIA: Grayscale

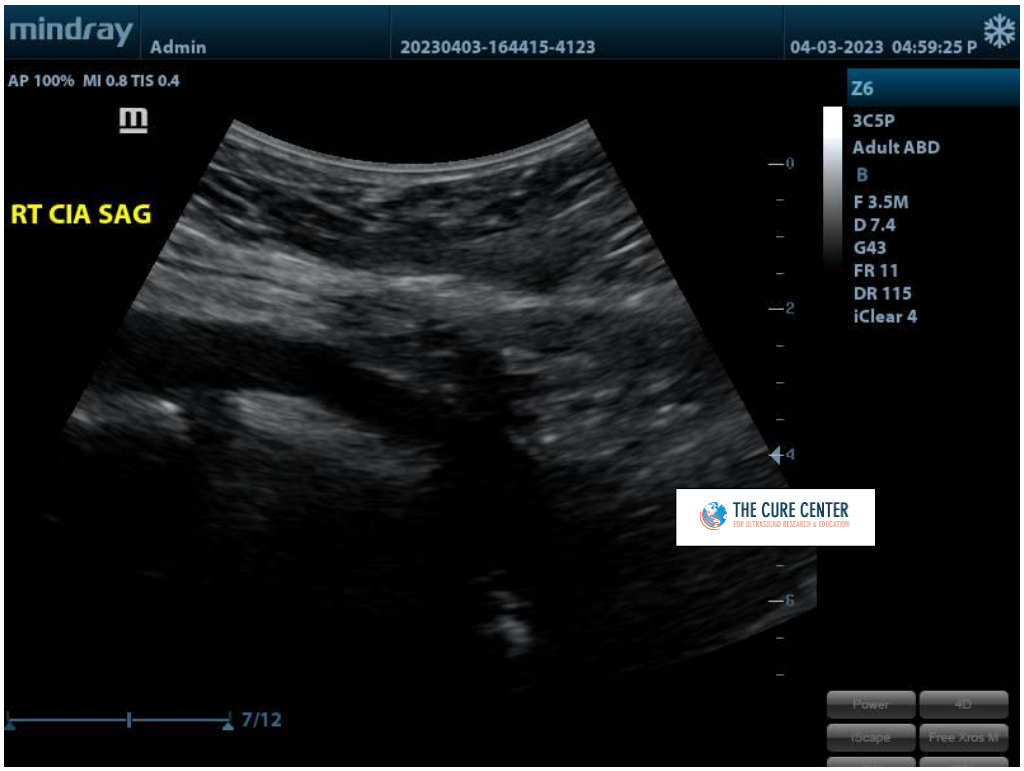


21. TRV RT CIA: W/ Measurement

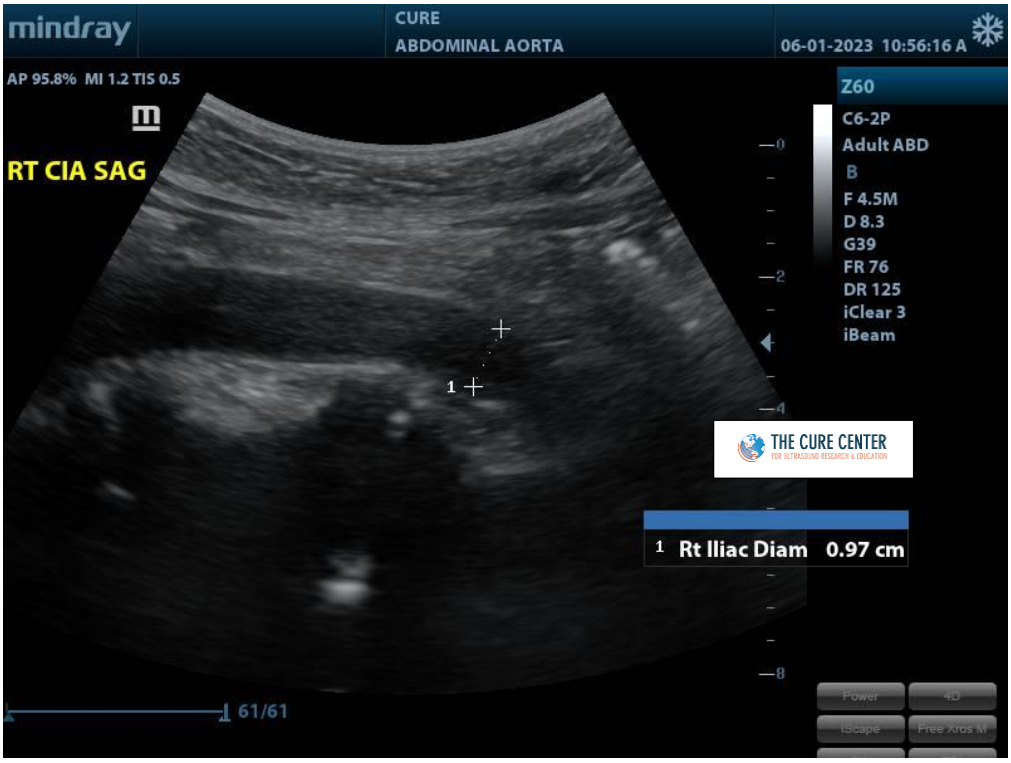


Sagittal RT CIA

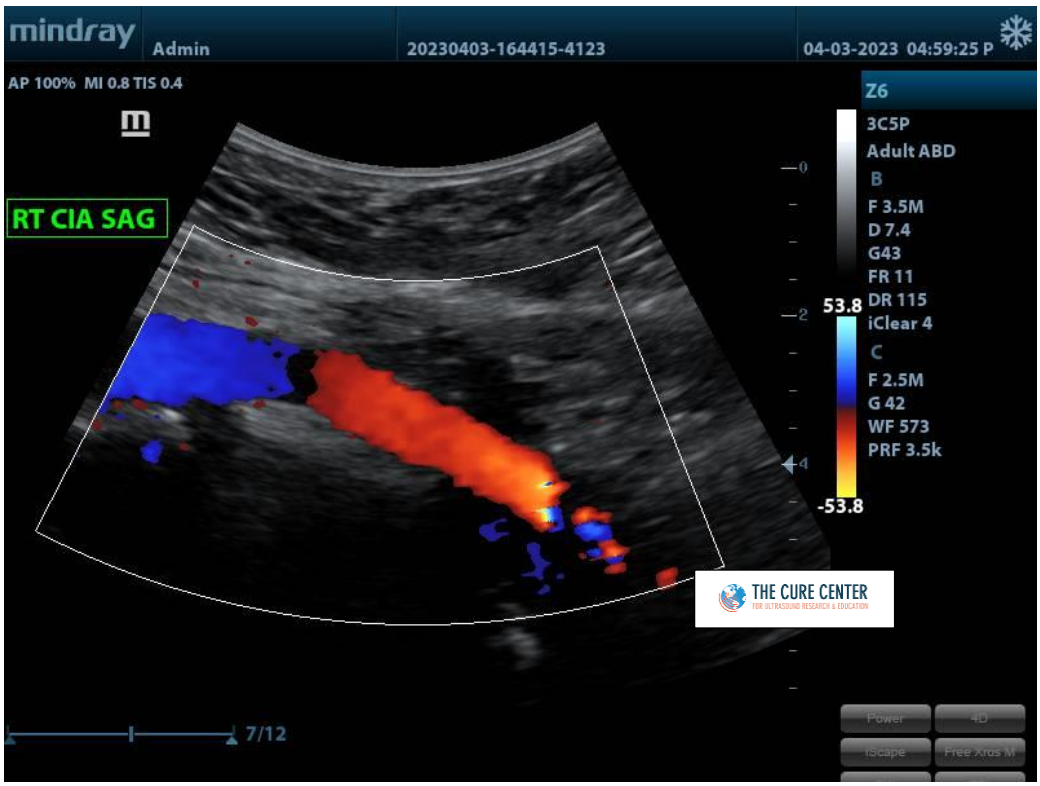
22. SAG RT CIA: Grayscale



23. SAG RT CIA: w/ Measurement



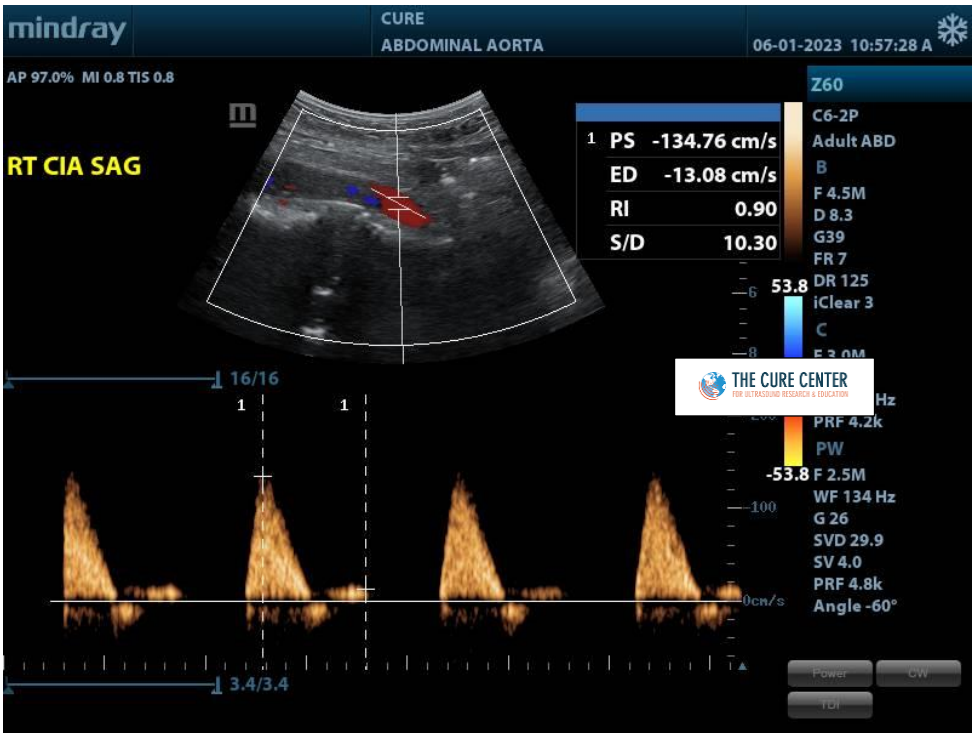
24. SAG RT CIA: w/ Color



25. SAG RT CIA: Doppler w/ PSV/ EDV Measurement

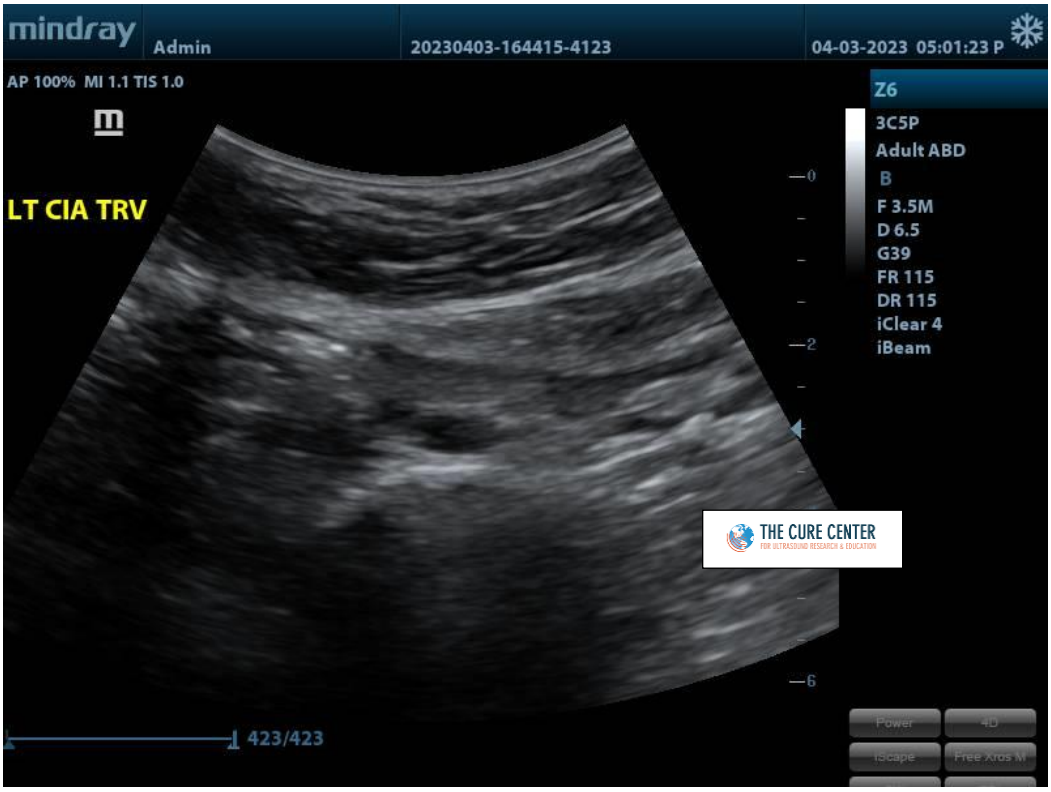
NOTE: Be sure to **angle** your sample gate to **60 degrees** for spectral Doppler images.

Remember to obtain measurements of the **PSV** and **EDV**, respectively.

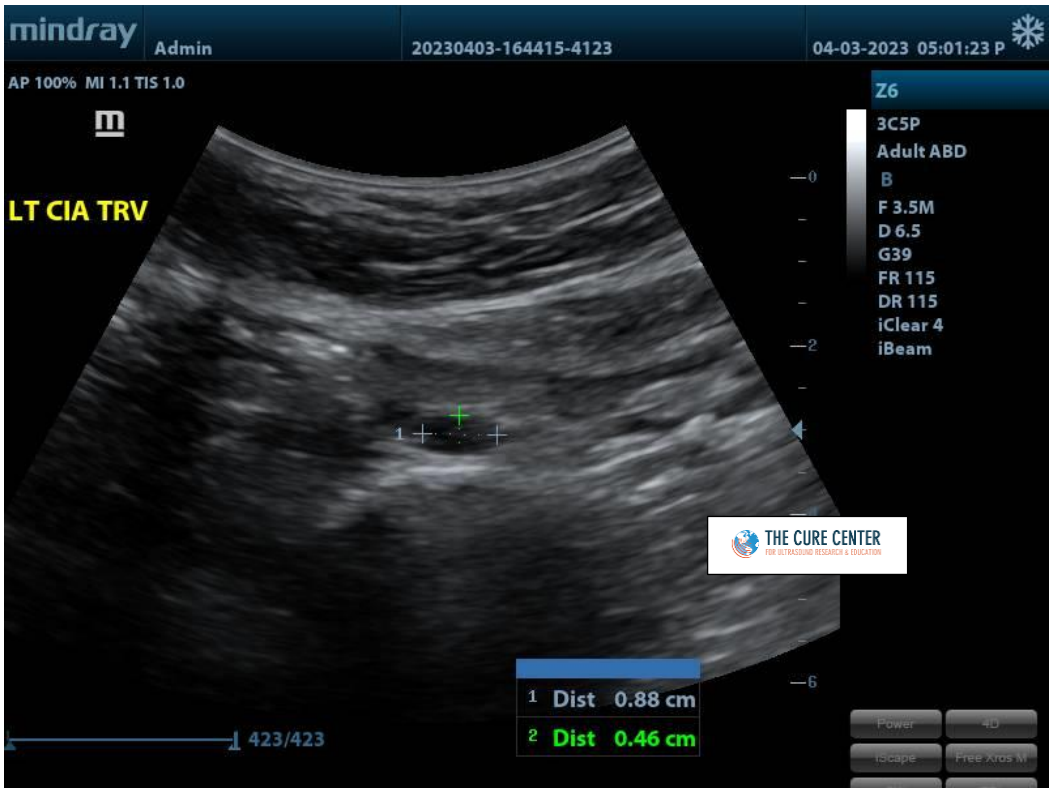


Transverse LT CIA

26. TRV LT CIA: Grayscale

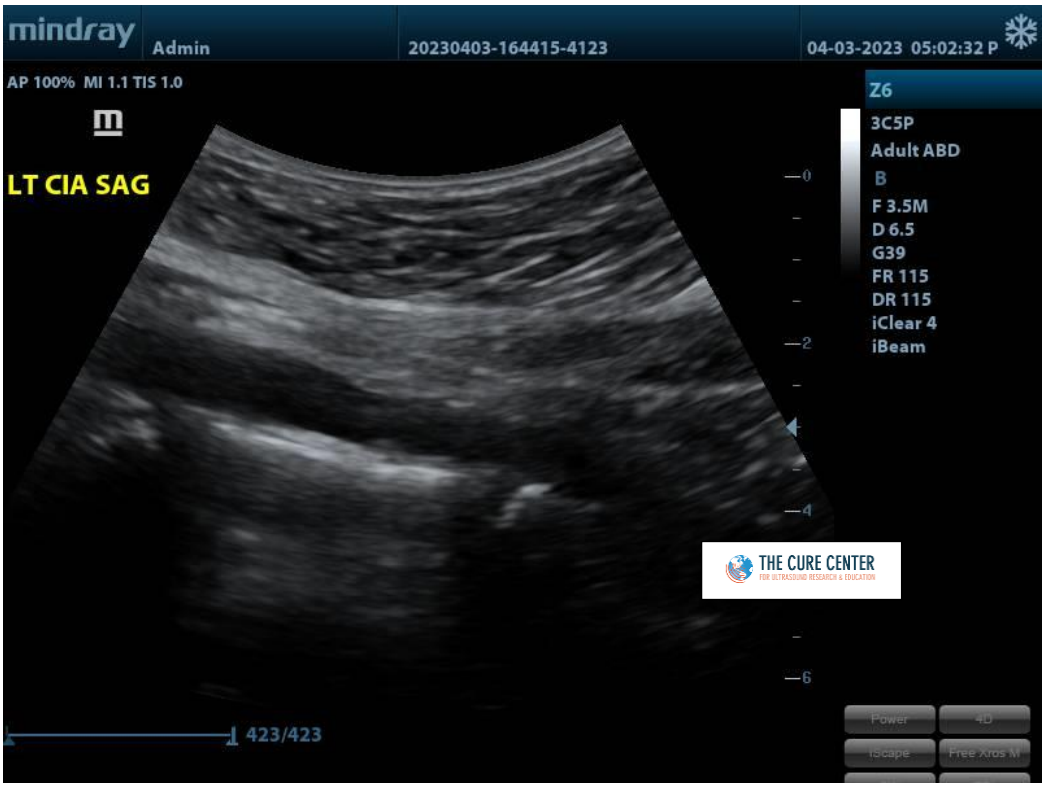


27. TRV LT CIA: w/ Measurement

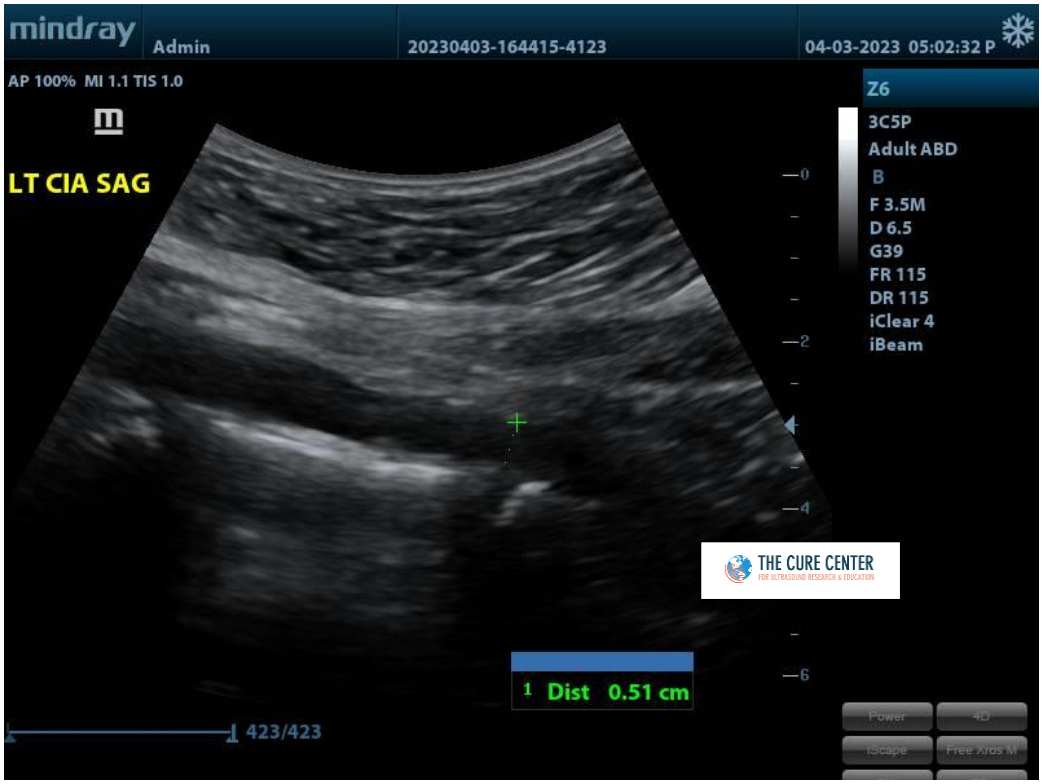


Sagittal LT CIA

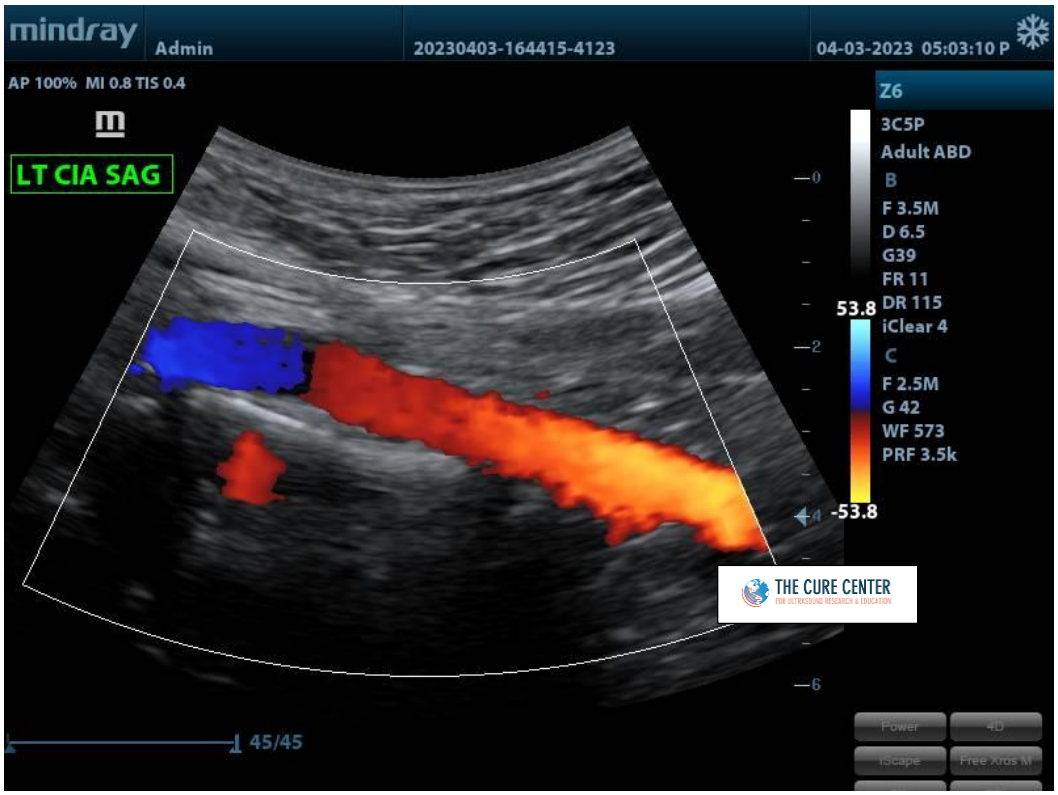
28. SAG LT CIA: Grayscale



29. SAG LT CIA: w/ Measurement



30. SAG LT CIA: w/ Color



31. SAG LT CIA: Doppler w/ PSV/ EDV Measurement



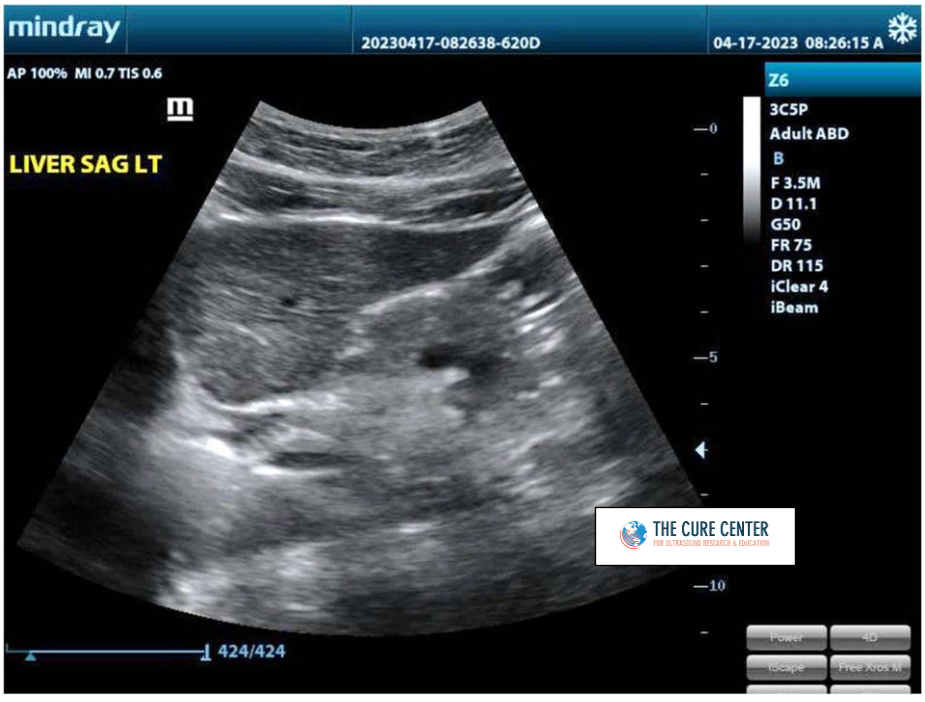
Liver Protocol

LIVER SAG LT:

1. SAG Left Lobe – Tip

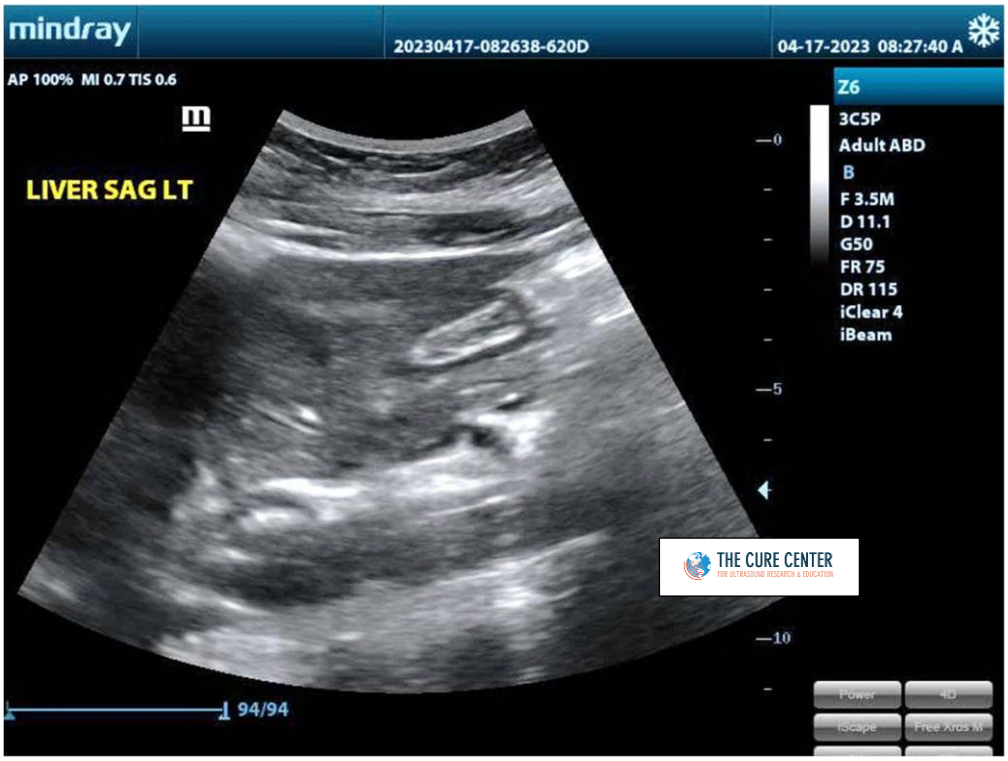
Note: your annotations should match the annotations in **blue (title above)**.

Note: Put the transducer in the middle of the stomach and angle out to the left of the patient's body.



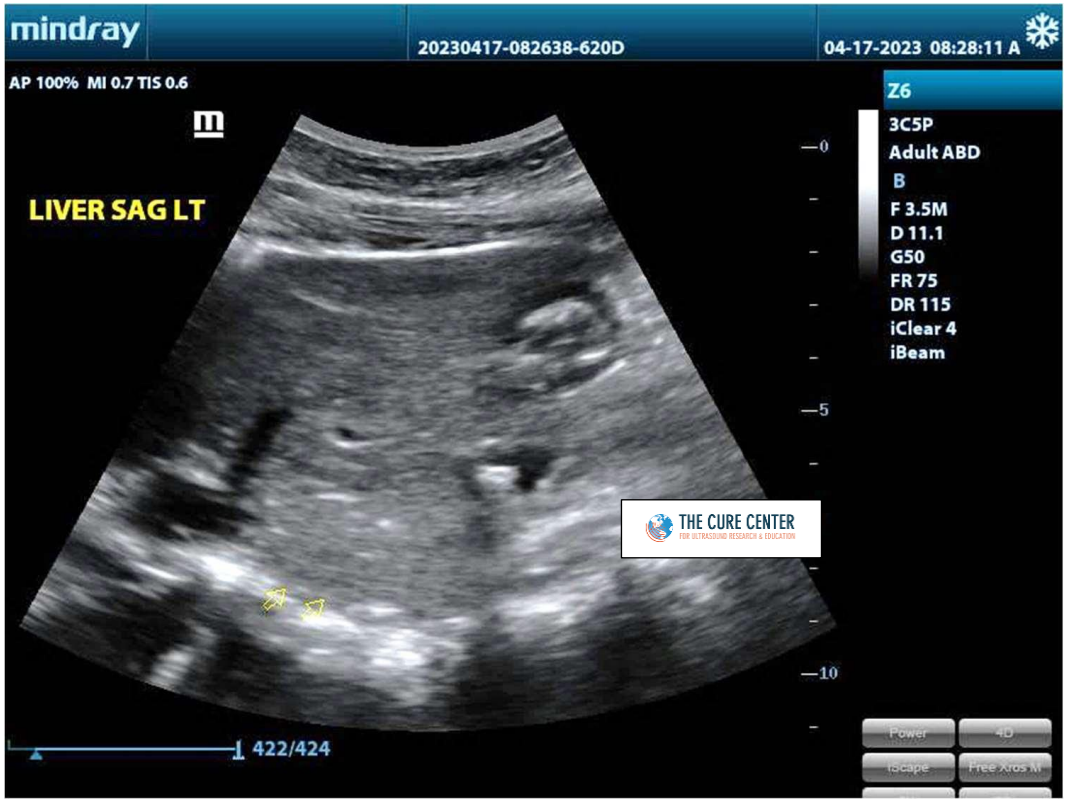
2. SAG Left Lobe – with the Aorta

Note: Stay where you are and start coming a little to the patient's right



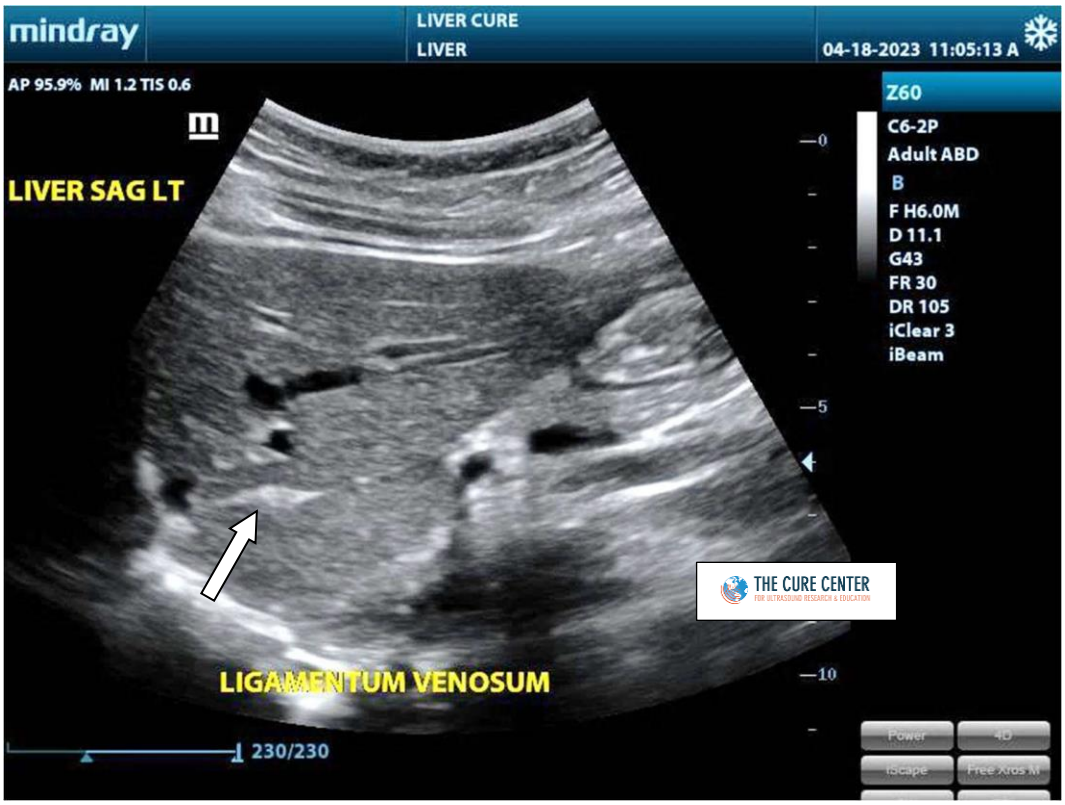
3. SAG Left Lobe – Caudate Lobe

Note: Stay where you are and come even more to the patient's right. Close the IVC and decrease the depth



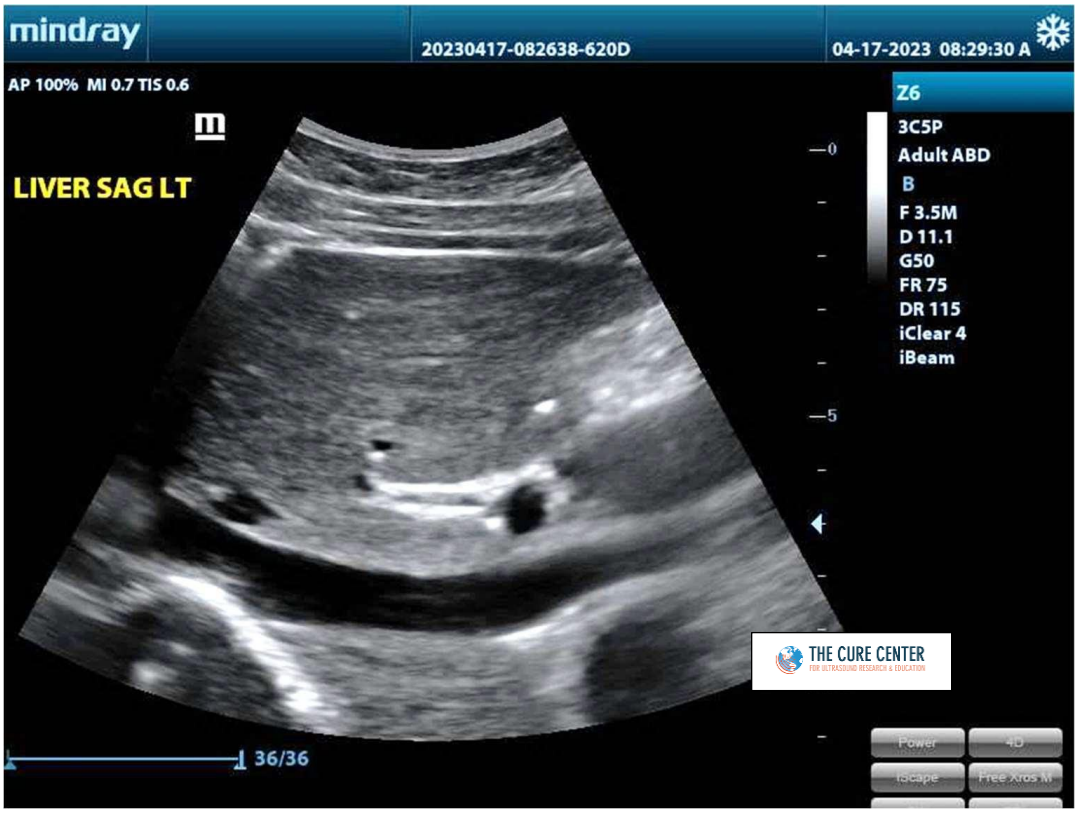
4. SAG Left Love – Ligamentum Venosum

Note: Close the Aorta and decrease the depth



5. SAG Left Lobe – with the Inferior Vena Cava (IVC)

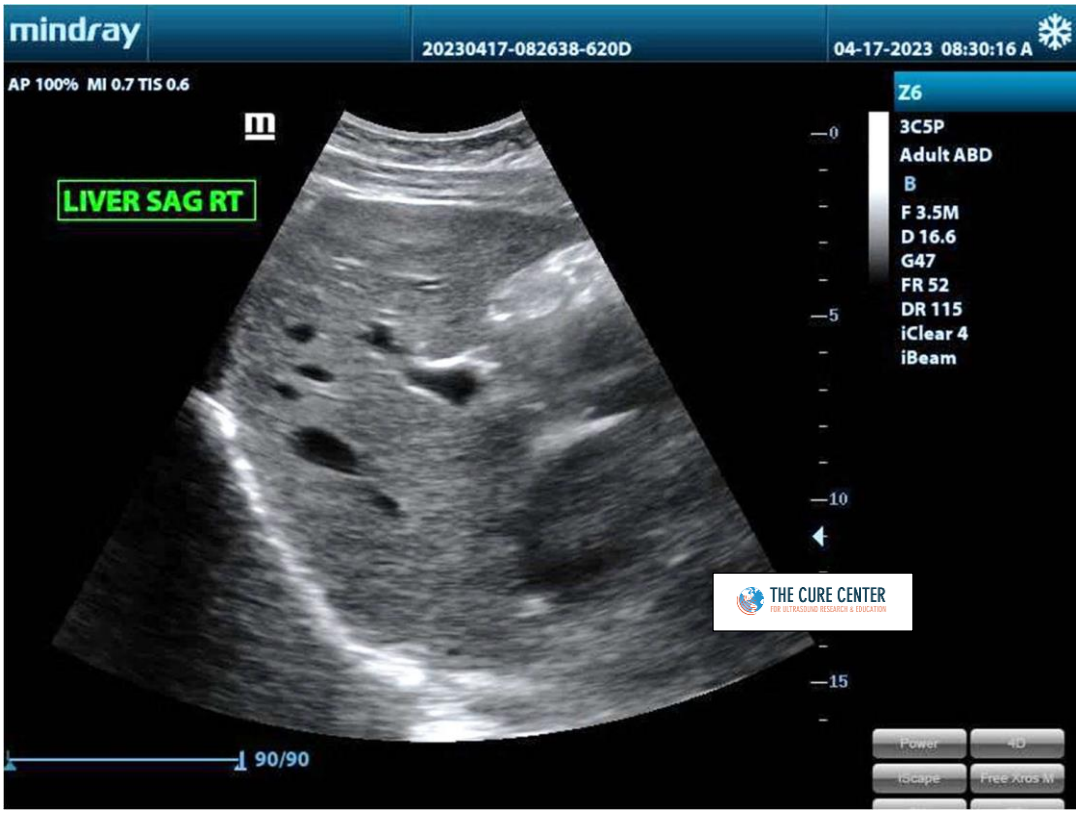
Note: Stay where you are and come even more to the patient's right until you see the heart moving. Decrease the depth



LIVER SAG RT:

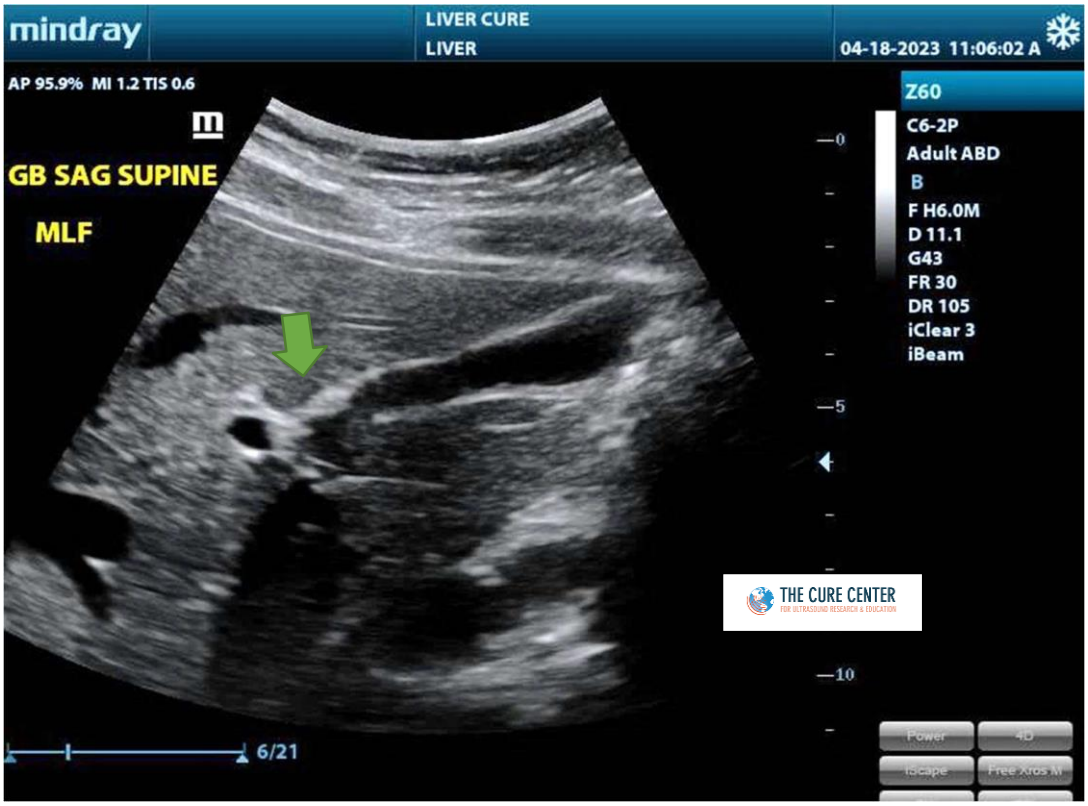
6. SAG Right Lobe - Long Liver

Note: Decrease the depth while you are going to the patient's right. You will obtain the left and right liver with some dome.



7. SAG Right Lobe – Main Lobar Fissure (MLF)

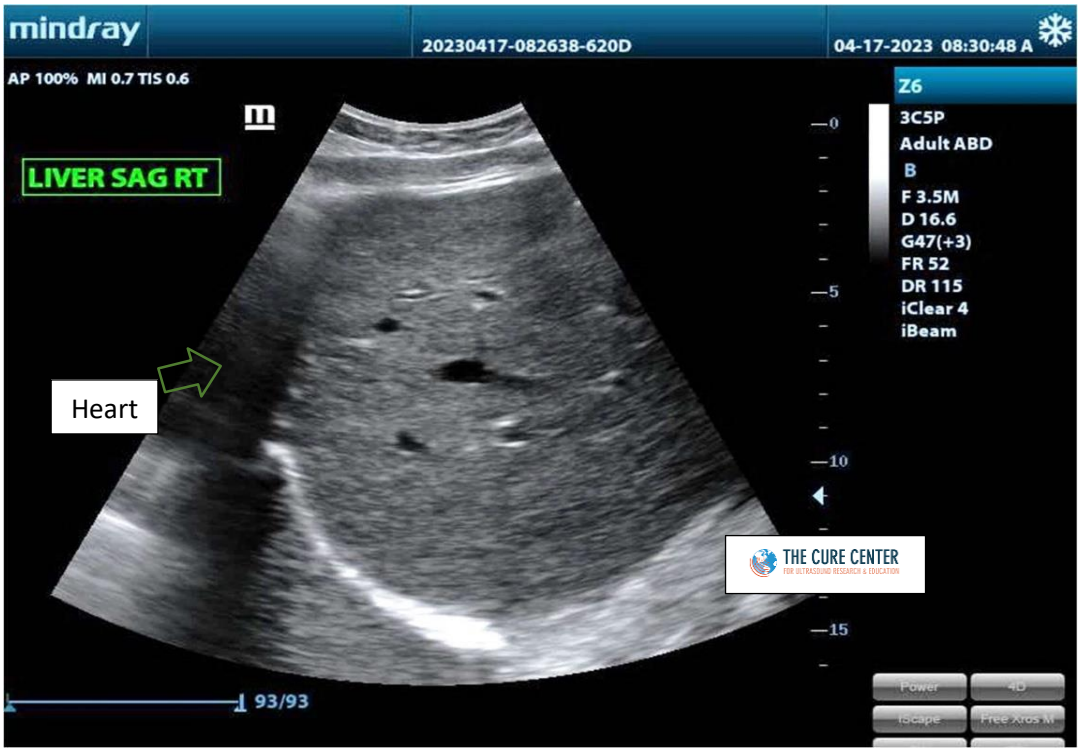
Note: Decrease the depth. Find the GB in the sag and connect it to the PV. The PV should look round.



FOR THE NEXT IMAGES 3,4, AND 5, ANGLE THE PROBE PERPENDICULAR TO THE DIRECTION OF THE RIGHT RIBS. SLIDE DOWN THE RIBS TO FIND 3, 4, AND 5.

8. SAG Right Lobe – Dome w/ Heart

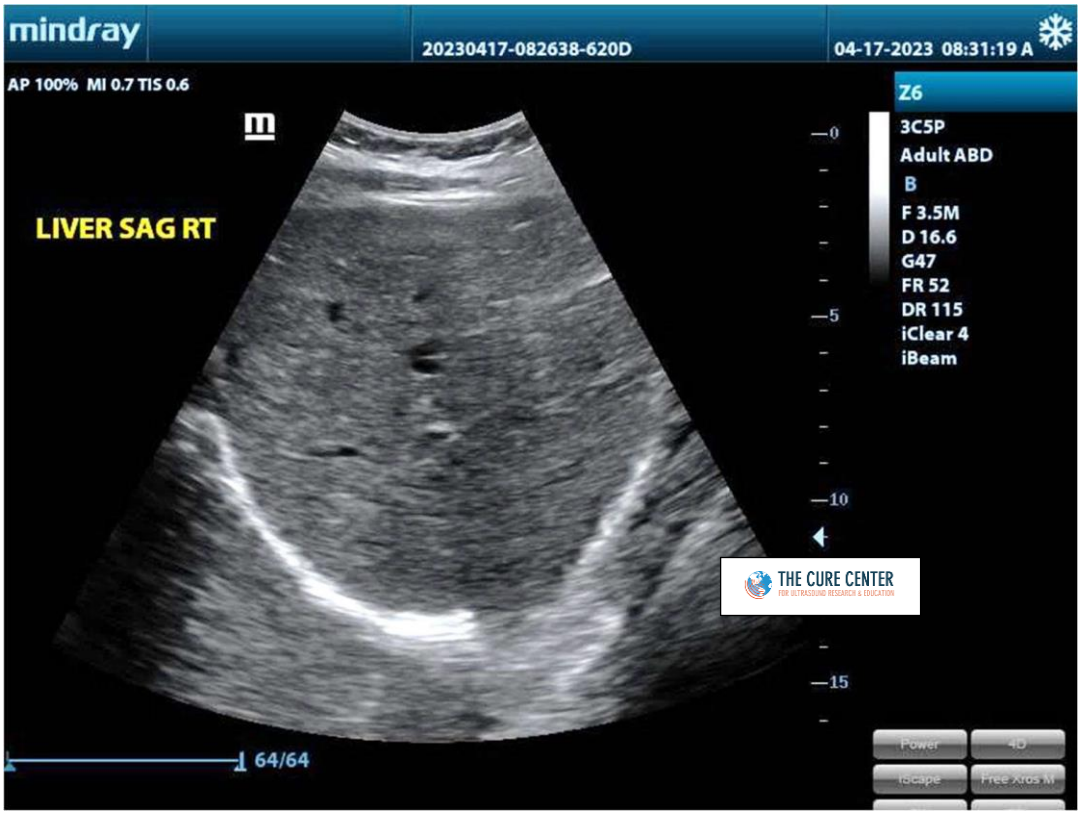
Note: Turn the patient away from you LLD for better images. Make sure the GB is not in the liver image. Close the dome of the liver to see the heart.



9. SAG Right Lobe – Dome Centered

Note: Turn the patient away from you LLD for better images. Make sure the GB is not in the liver image.

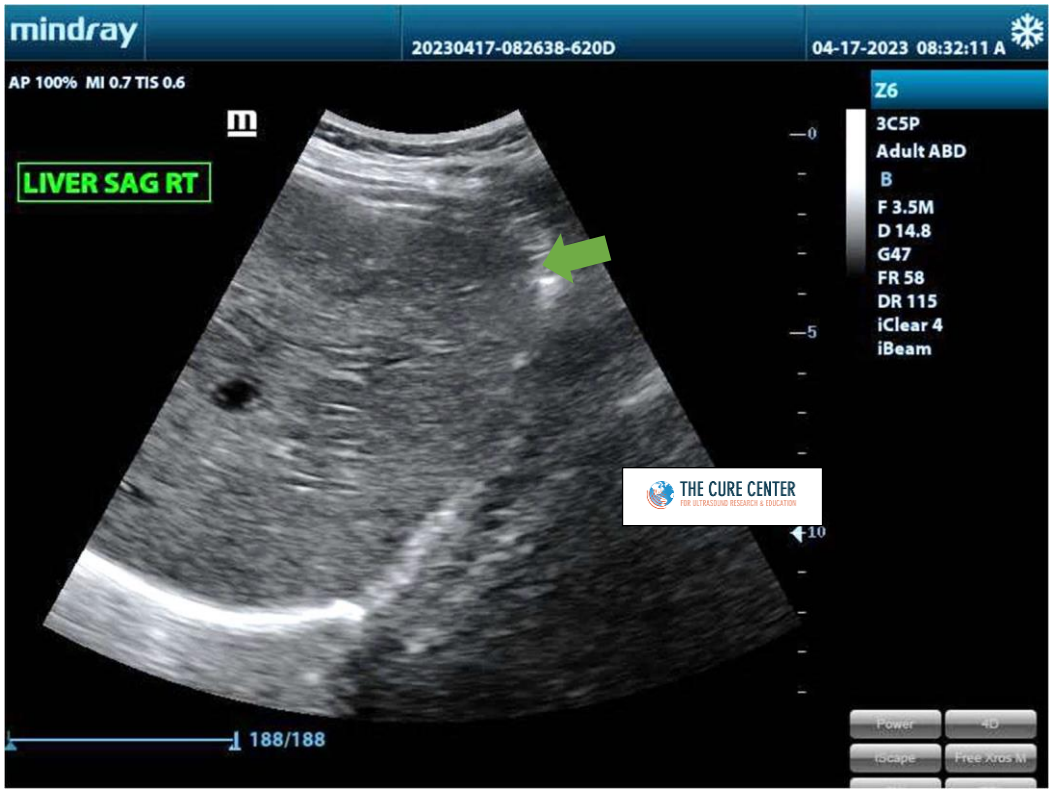
Center your liver to see a complete view



10. SAG Right Lobe – Dome w/ Corner

Note: Turn the patient away from you LLD for better images. Make sure the GB is not in the liver image.

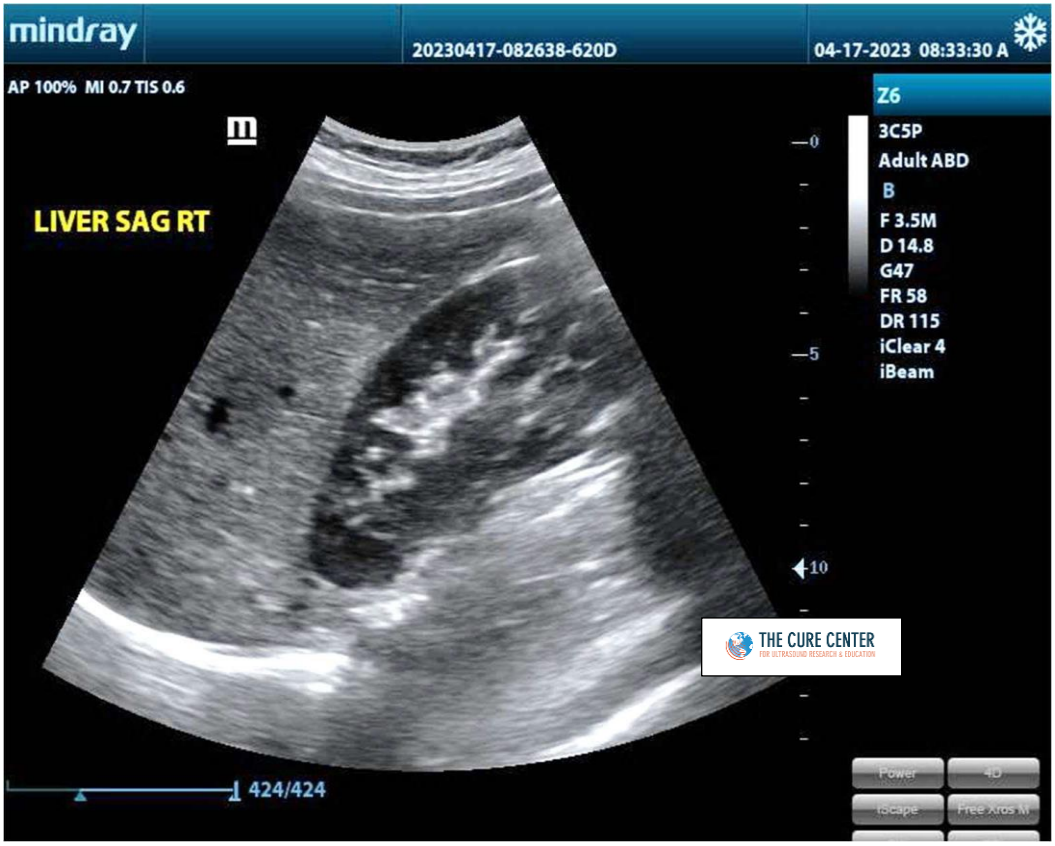
Slide down more inferior to see the other corner of the liver.



11. SAG Right Lobe – Liver/ Kidney Interface

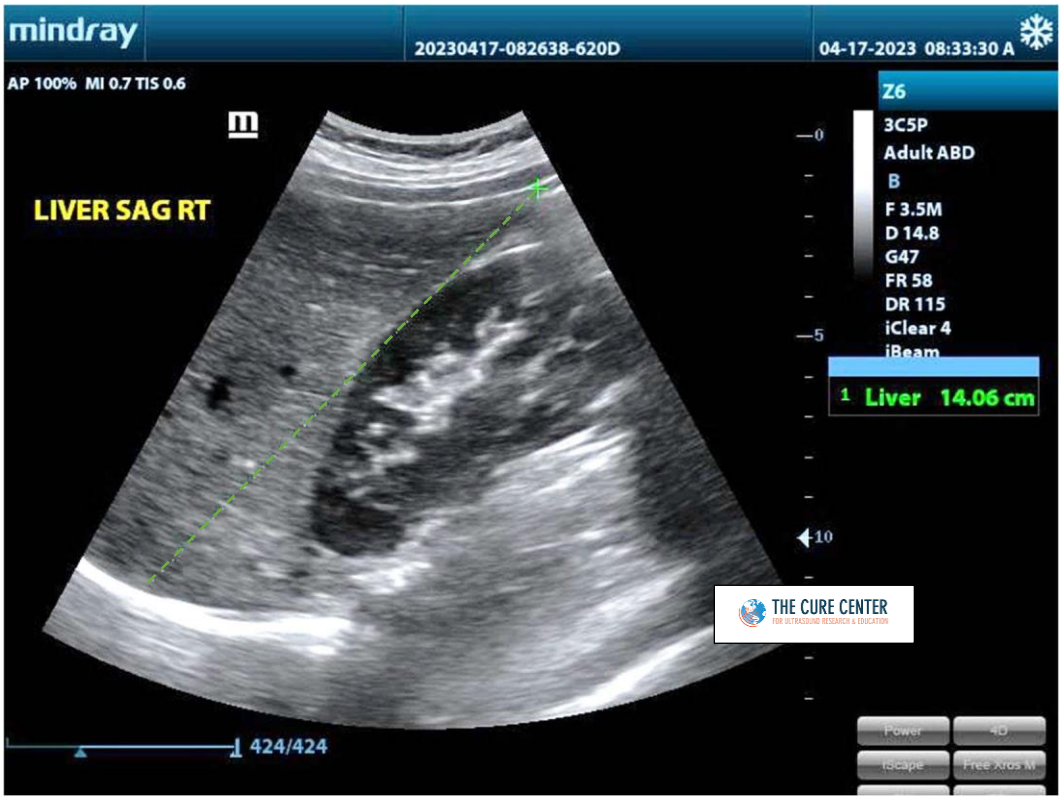
Note: Place the transducer inferior to the right ribs, then turn the notch toward the patient's head.

Make sure you see the tip of the liver to measure the liver.



12. SAG Right Lobe – Liver/ Kidney Interface w/ measurement

Note: Stay where you are and measure the liver from the tip to the dome inner to inner.



LIVER TRV LT:

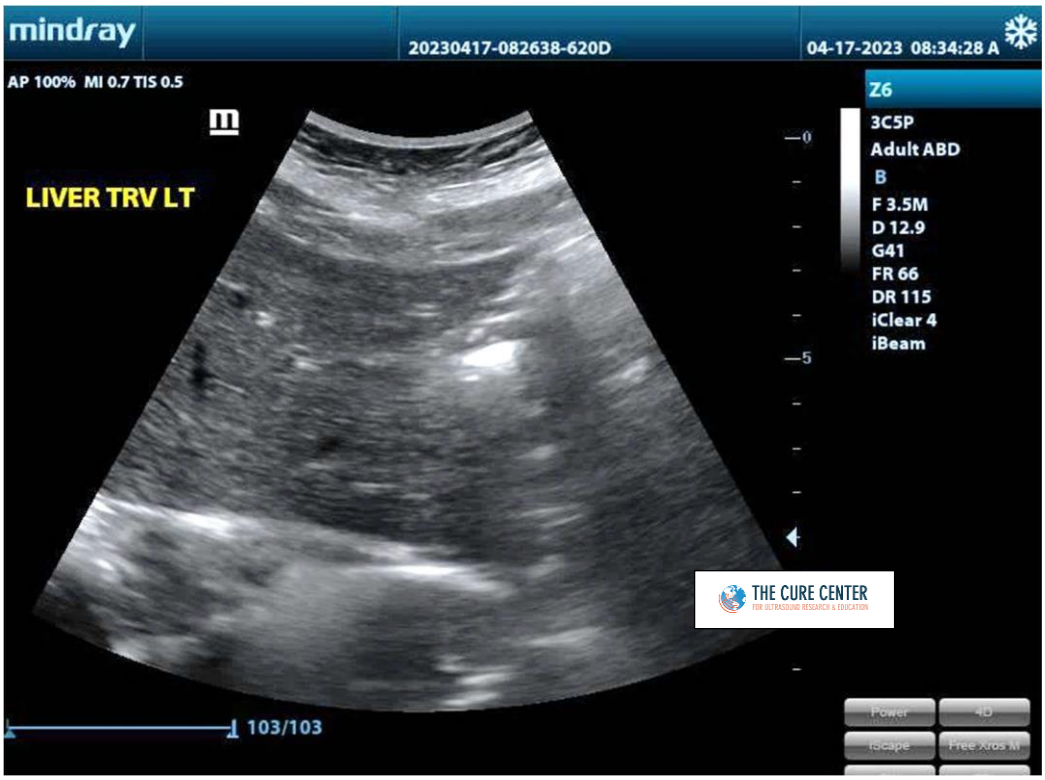
13. TRV Left Lobe – Ligamentum Teres

Note: Place the transducer toward the left chest. Keep going up and down on the left lobe to find the echogenic round spot.



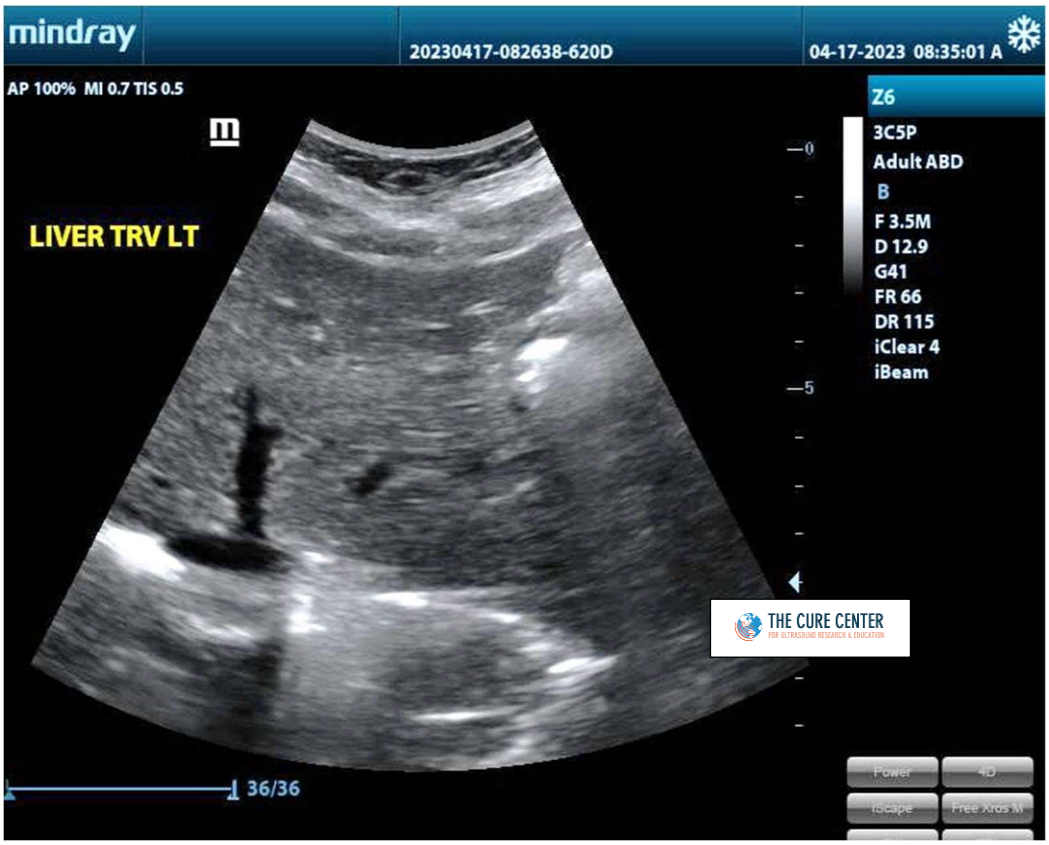
14. TRV Left Lobe – Tip

Note: Stay where you are and angle transducer up towards the chest. The liver Left Tip should look triangular.



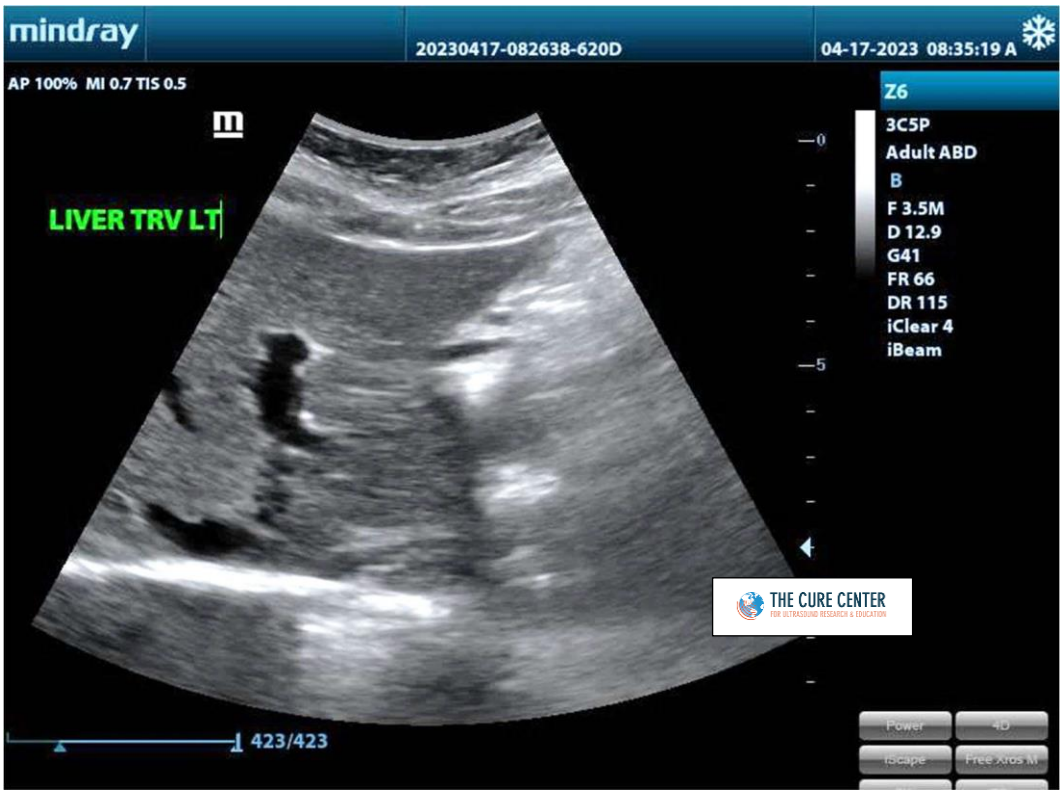
15.TRV Left Lobe – Left Hepatic Vein (LHV)

Note: Move the transducer a little down to find the left hepatic vein coming off the IVC.



16. TRV Left Lobe – Left Portal Vein (LPV)

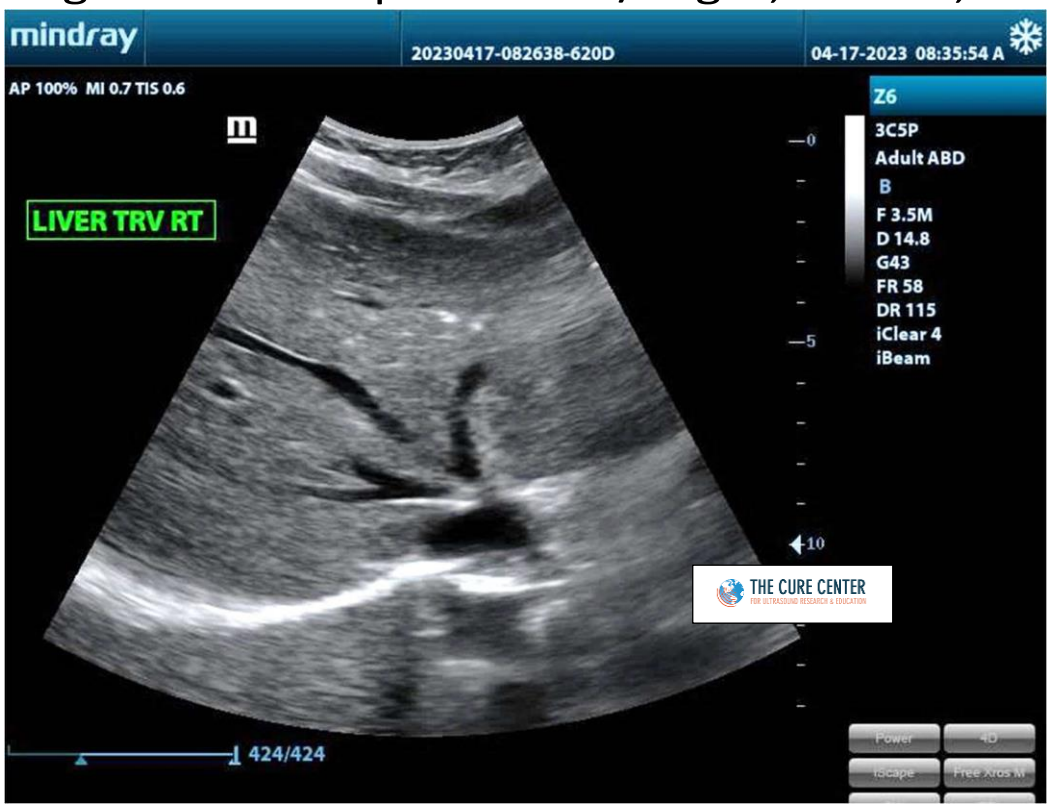
Note: Move the transducer further down to find the left portal vein.



LIVER TRV RT:

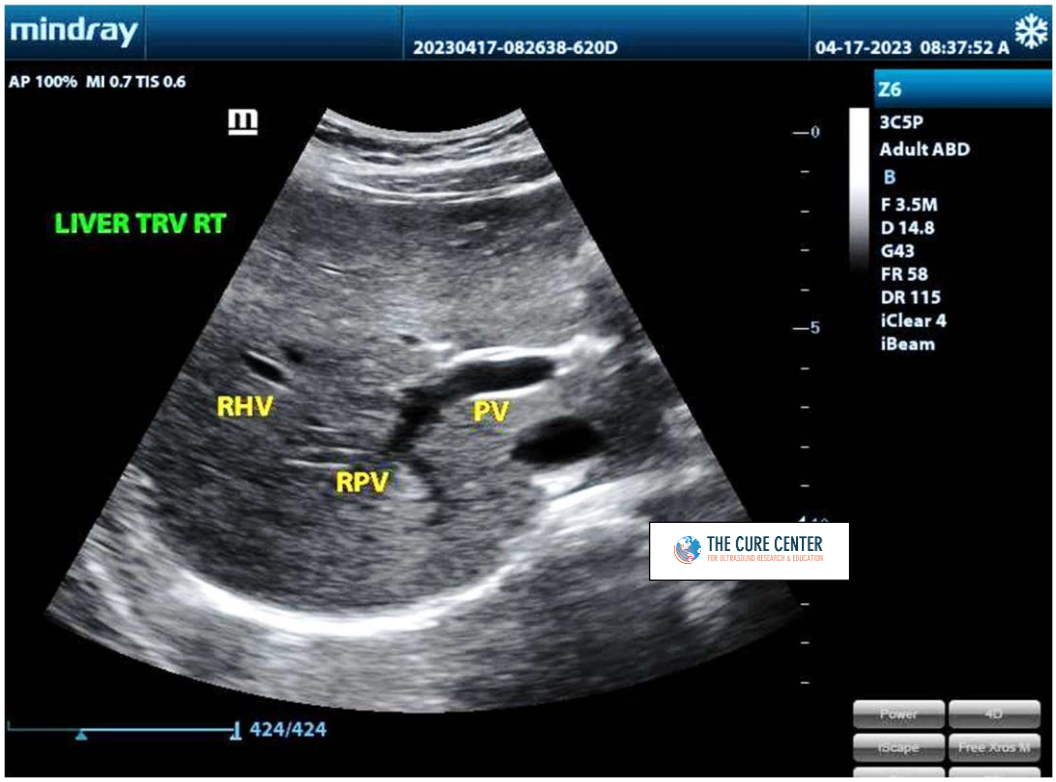
17. TRV Right Lobe – Hepatic Veins/ Right, Middle, Left

Note: angle the transducer towards the right rib (notch is facing the right) to find the hepatic veins. They all have to be coming off the IVC (sometimes you will see only two HV out of three).



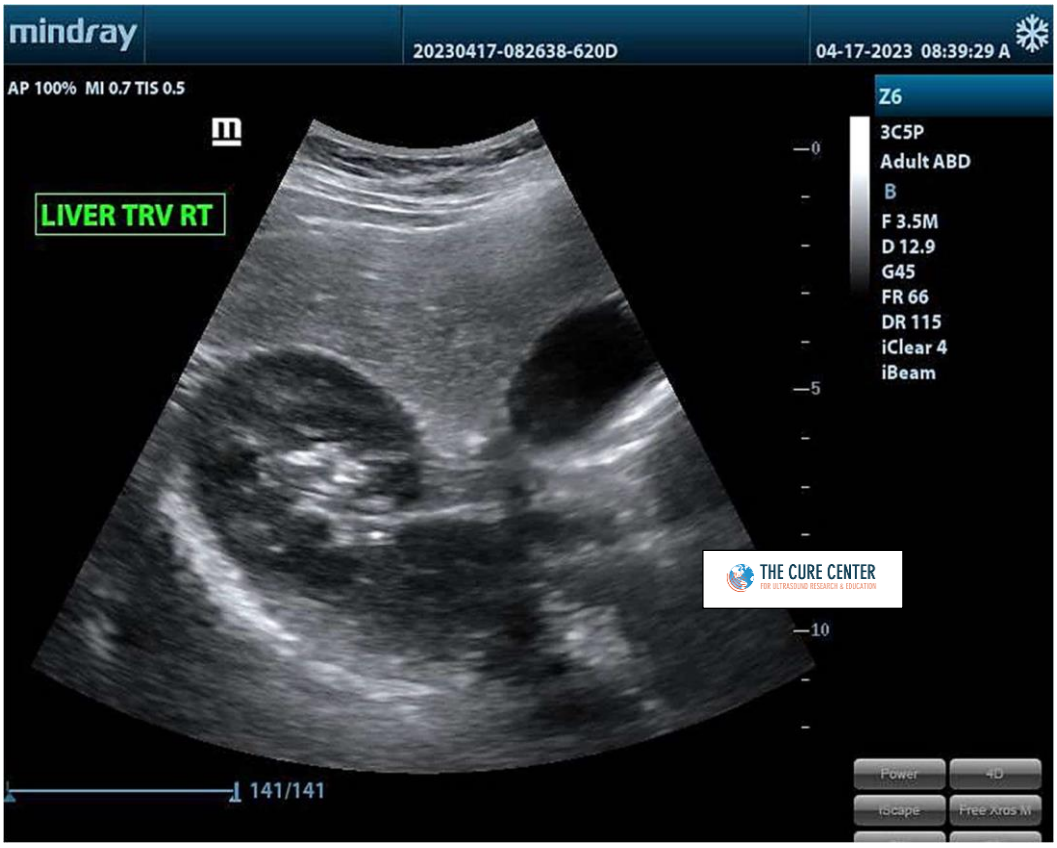
18. TRV Right Lobe – RHV/ RPV/ PV

Note: Place the transducer's notch toward you. The probe needs to be perpendicular to the right ribs.



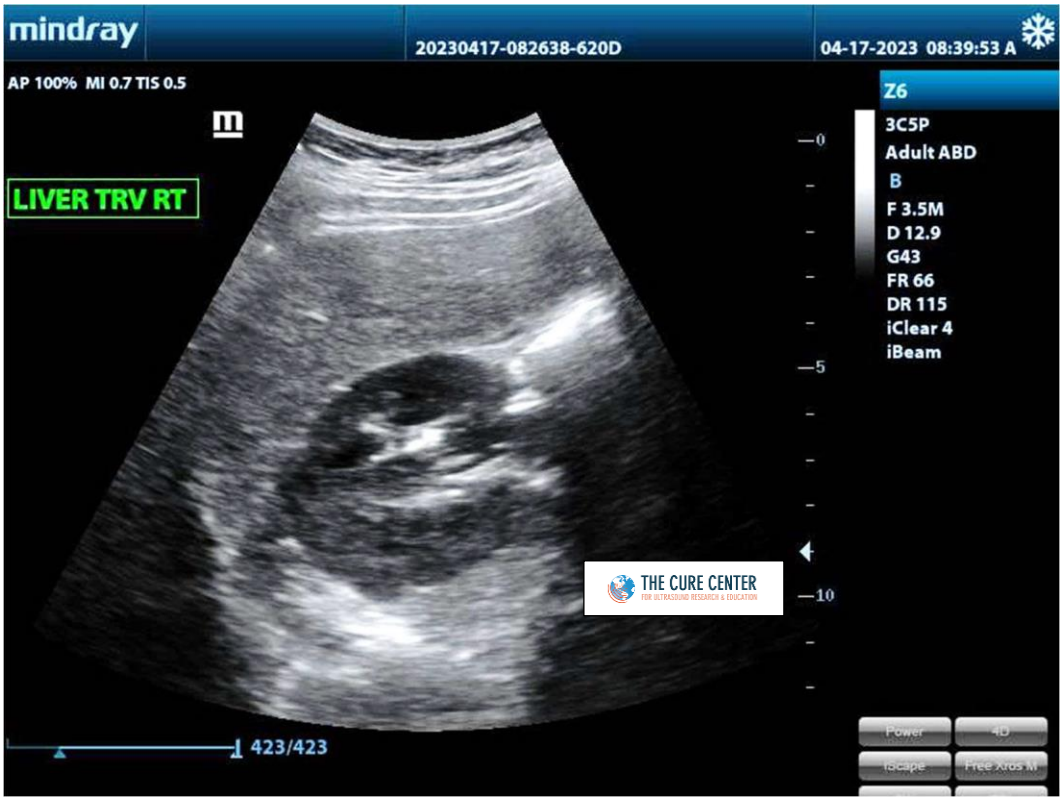
19.TRV Right Lobe – Liver/ Kidney/GB

Note: Stay where you are and slide the prove further down inferior to see the GB, when you find it, twist your transducer to get the right kidney in the picture at the same time



20. TRV Right Lobe – Liver/Kid

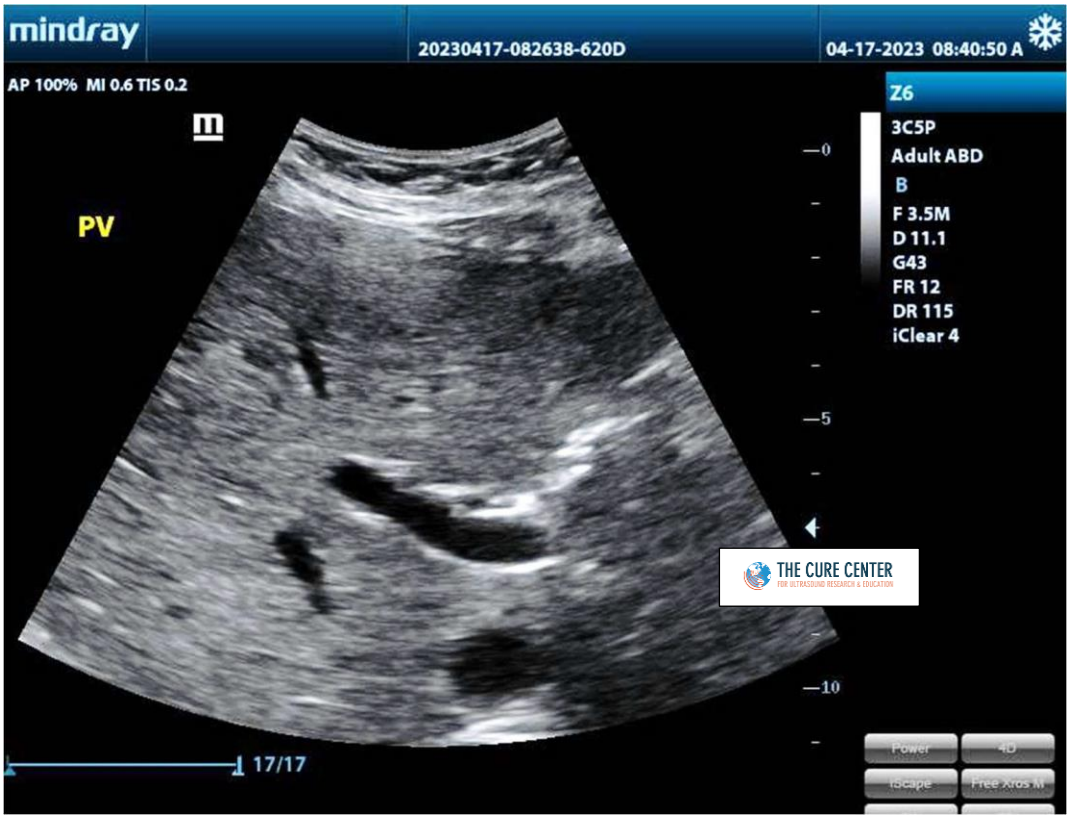
Note: Slide down more inferiorly and remove the GB from your picture to get this image.



21. TRV Right Lobe – PV No Color

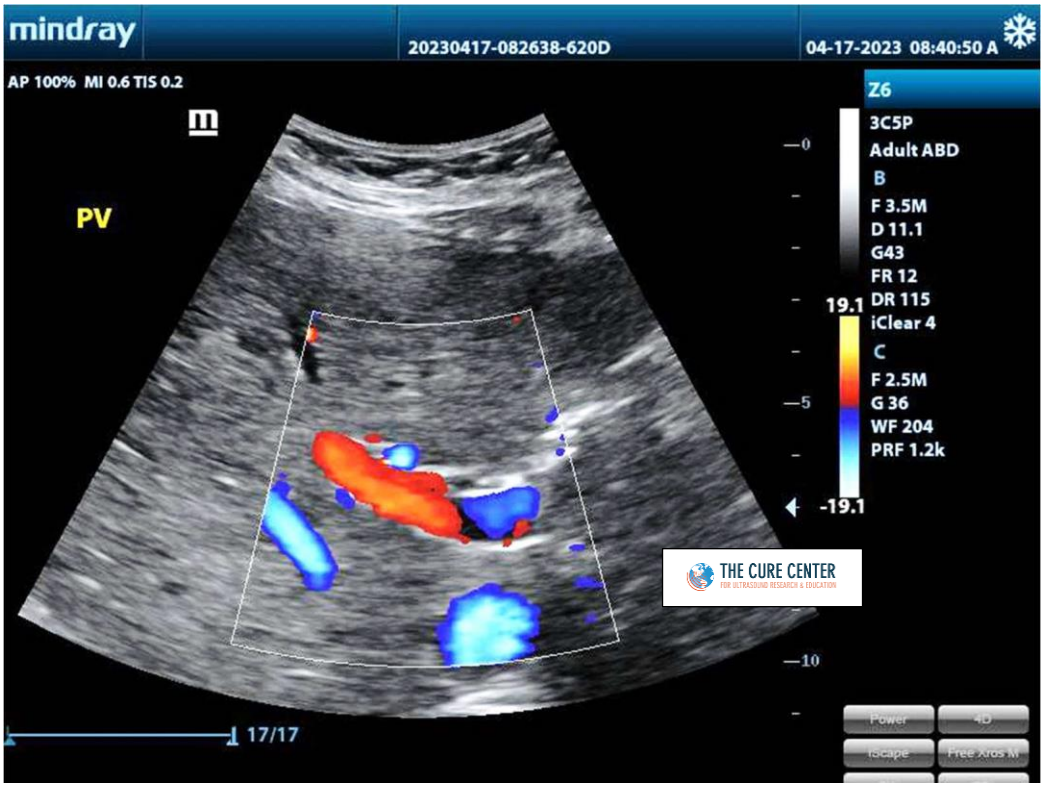
Note: Your transducer will be transverse while the portal vein is elongated.

Note: Make sure that half of the prove is on the right ribs and other half on the patient's belly. PV needs to look red, also on the US the doppler spectral waveform is above baseline.



22. TRV Right Lobe – PV w/ Color

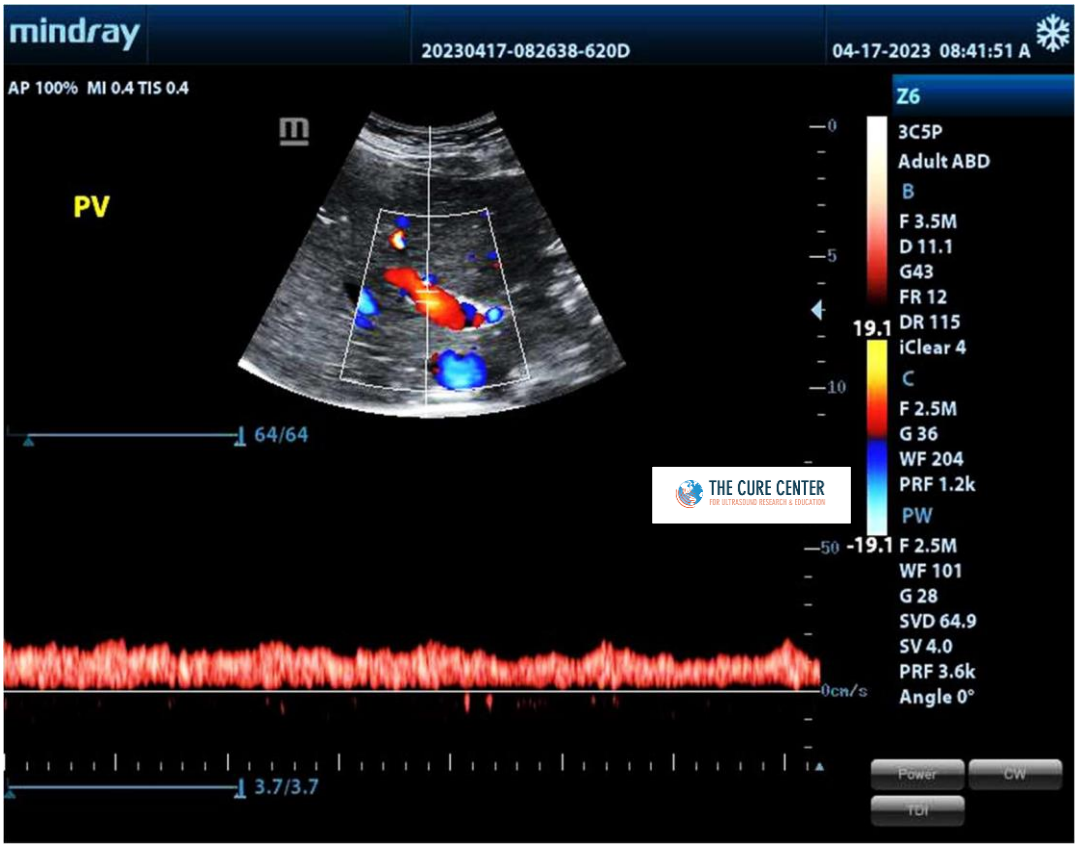
Note: Make sure that half of the prove is on the right ribs and other half on the patient's belly. PV needs to look red, also on the US the doppler spectral waveform is above baseline.



23. TRV Right Lobe – PV Color Doppler

Note: Your transducer will be transverse while the portal vein is elongated. The Doppler needs to be straight down. **No angle is needed** with the portal vein.

Note: Make sure that half of the prove is on the right ribs and other half on the patient's belly. PV needs to look red, also on the US the doppler spectral waveform is above baseline.



The Gallbladder and the Biliary Tree Protocol

Each set of Sag & TRV images must be done when the patient is:

Supine, LLD, Upright

SUPINE:

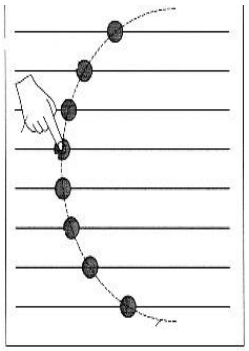
1. SAG GB - Fundus/ Body/ Neck

Note: Your annotations should match the labels within each ultrasound image.

Show the fundus, the body, and the

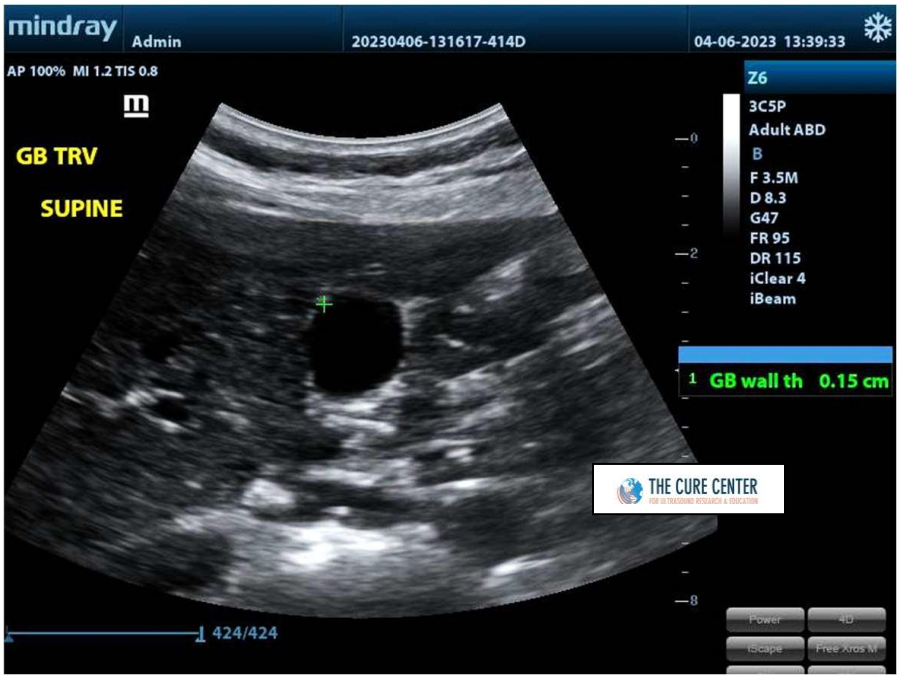


Note: TGC Placement Should be half-moon.



2. TRV GB – GB w/ Measurement

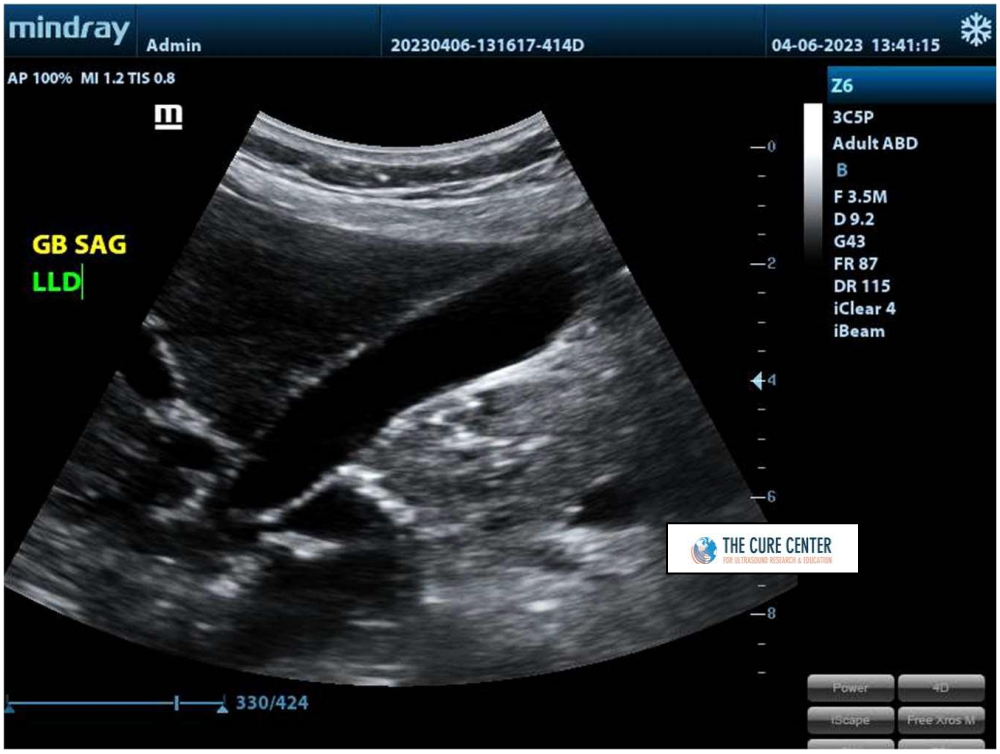
Note: Measure the GB anterior wall (the echogenic part only). You need one image of measurements in any view **AP**.



LLD:

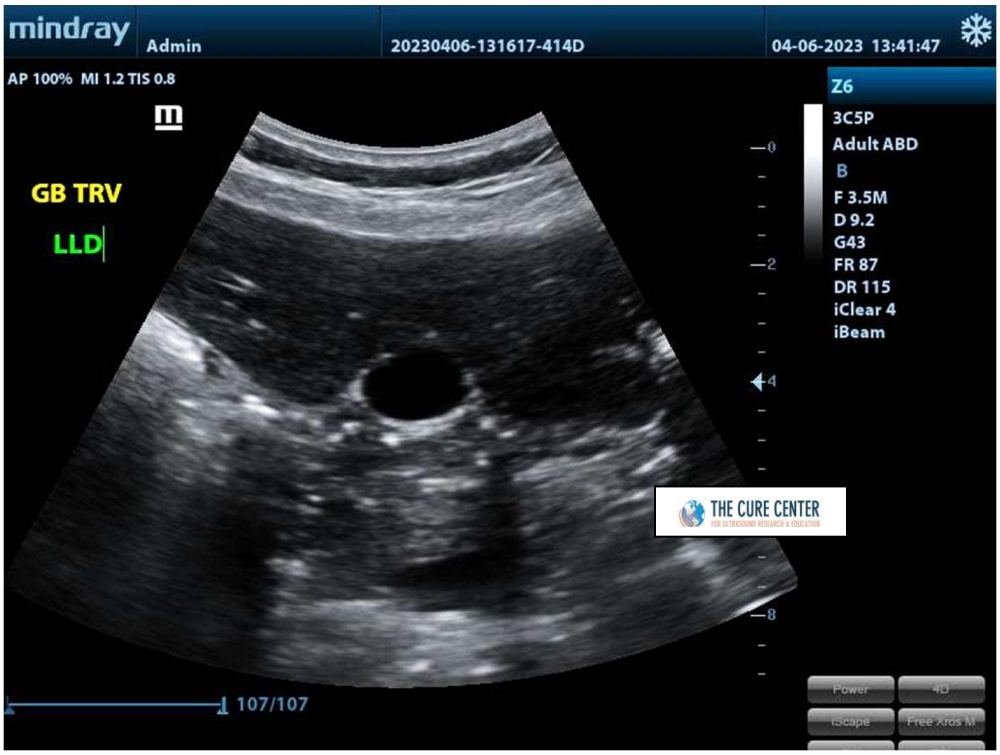
3. SAG GB - Fundus/ Body/ Neck

Note: Show the fundus, the body, and the neck. Make sure your GB in the center of the image



4. TRV GB – GB

Note: Turn on the GB in TRV. Make sure you center your images at all



UPRIGHT:

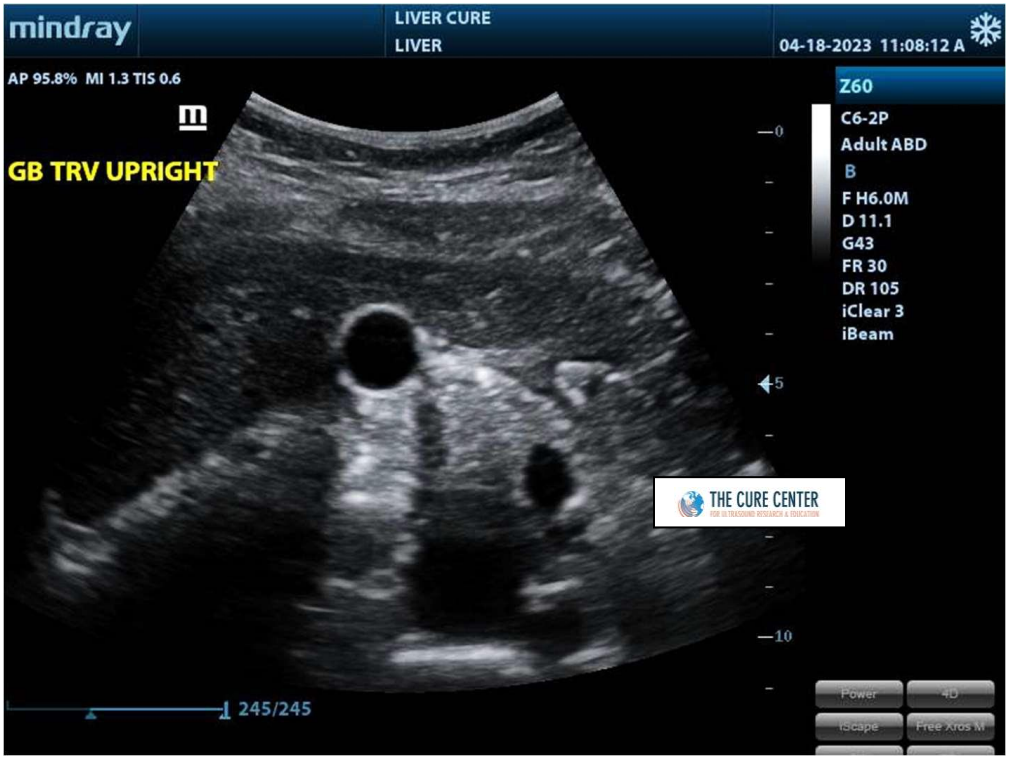
5. SAG GB - Fundus/ Body/ Neck

Note: Show the fundus, the body, and the neck. Make sure your GB in the center of the image



6. TRV GB – GB

Note: Turn on the GB in TRV. Make sure you center your images at all



7. SAG CBD / PV

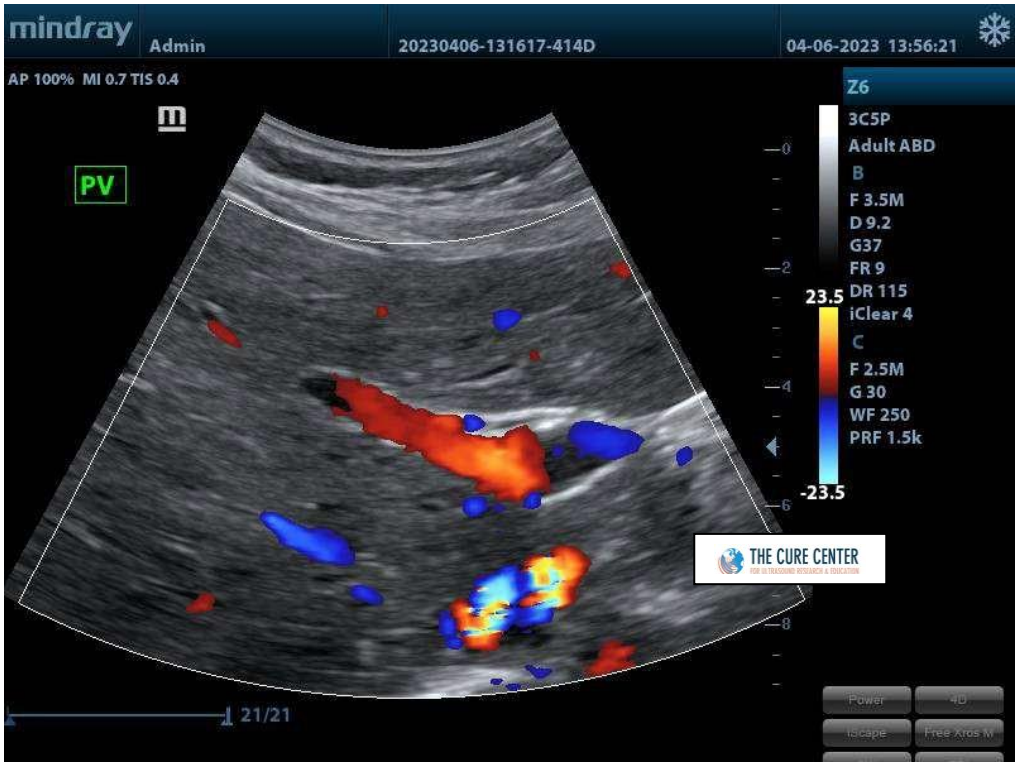
Note: Your transducer needs to be in TRV while your image is elongated. Put the probe half on the RT chest and the other half on the belly. The CBD is parallel to the PV.



8. SAG CBD/ PV – Color

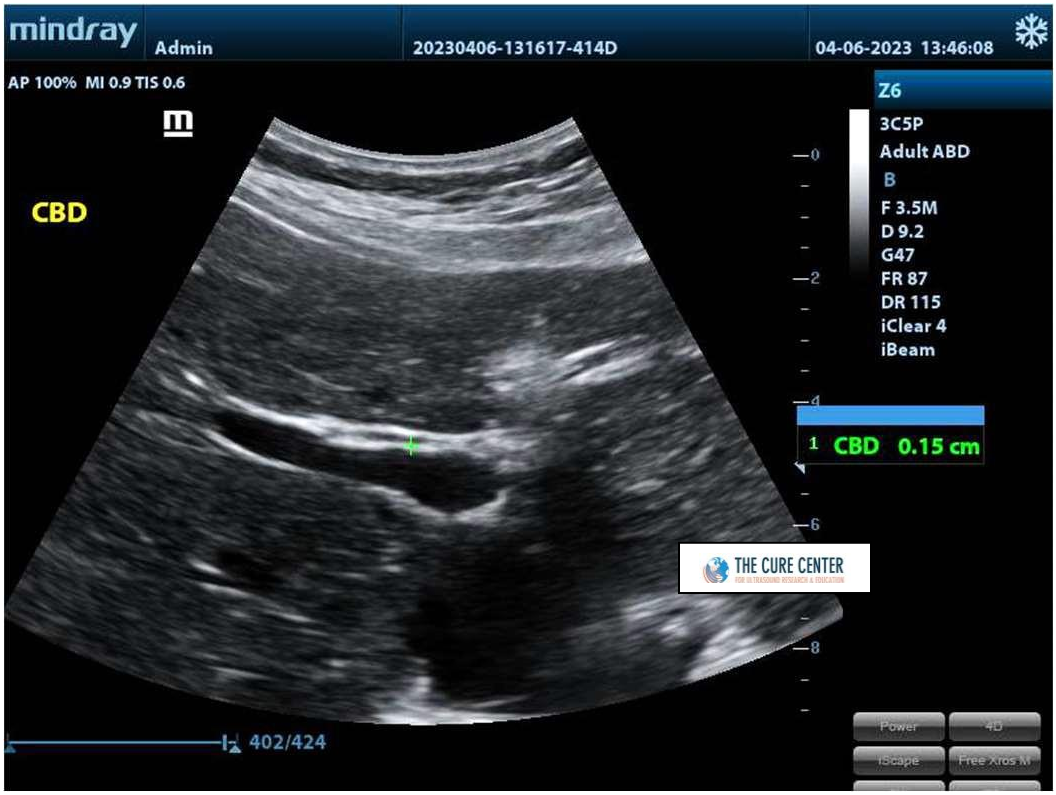
Note: In the same image, add color. Your color box should cover the image you are taking.

Remember, the CBD does not take color.



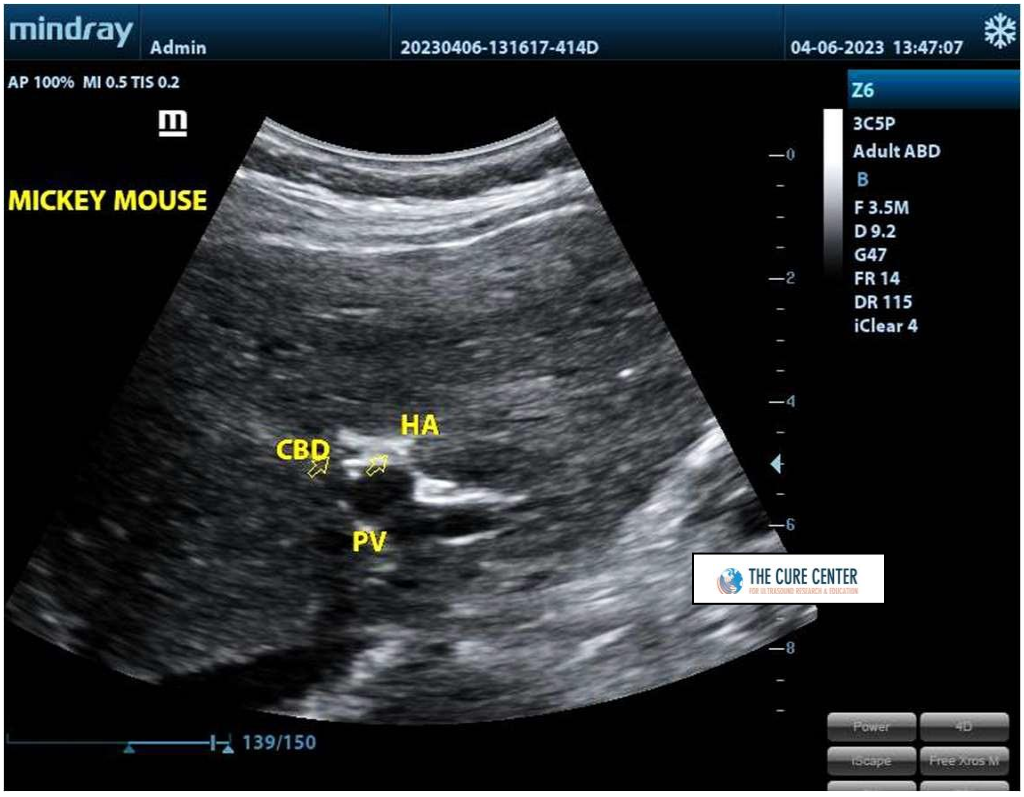
9. SAG CBD/ PV – CBD Measurement

Note: In the same image, take the color Doppler off. Measure the CBD **AP** (the anechoic portion only).



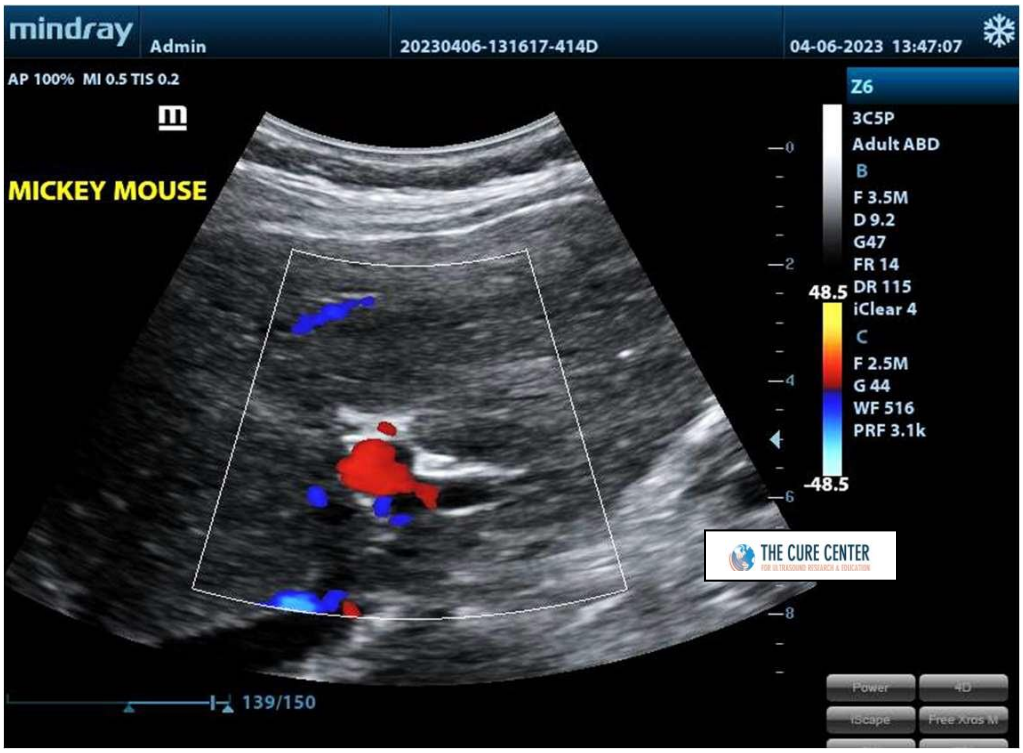
10. TRV PORTAL TRIAD – Grayscale

Note: Turn on the same image above TRV. You will get the Mickey Mouse Sign. PV is the head, then two ears (CBD and HA).



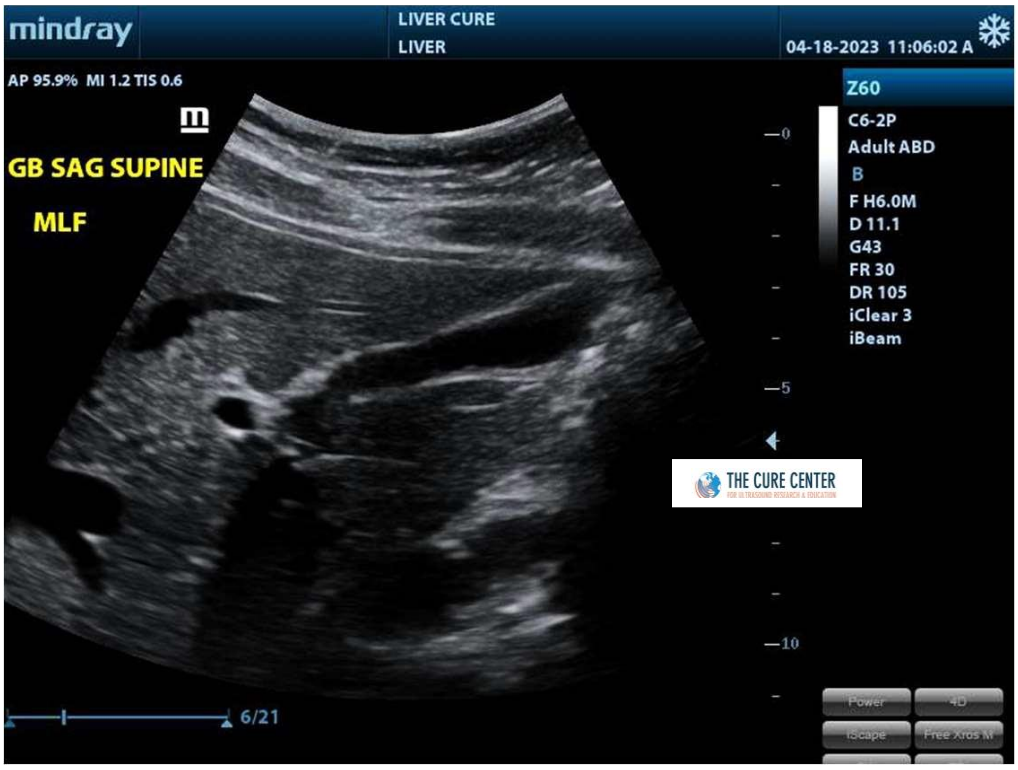
11. TRV PORTAL TRIAD – Color

Note: In the same image, add color Doppler. The HA and the PV will fill up with color, the CBD will not.



14. TRV GB MLF - Main Lobar Fissure (MLF)

Note: Elongate the GB in sagittal and try to connect it to the RPV (should look round). The echogenic line between them is the MLF. Pick the position that is best/easier for you.

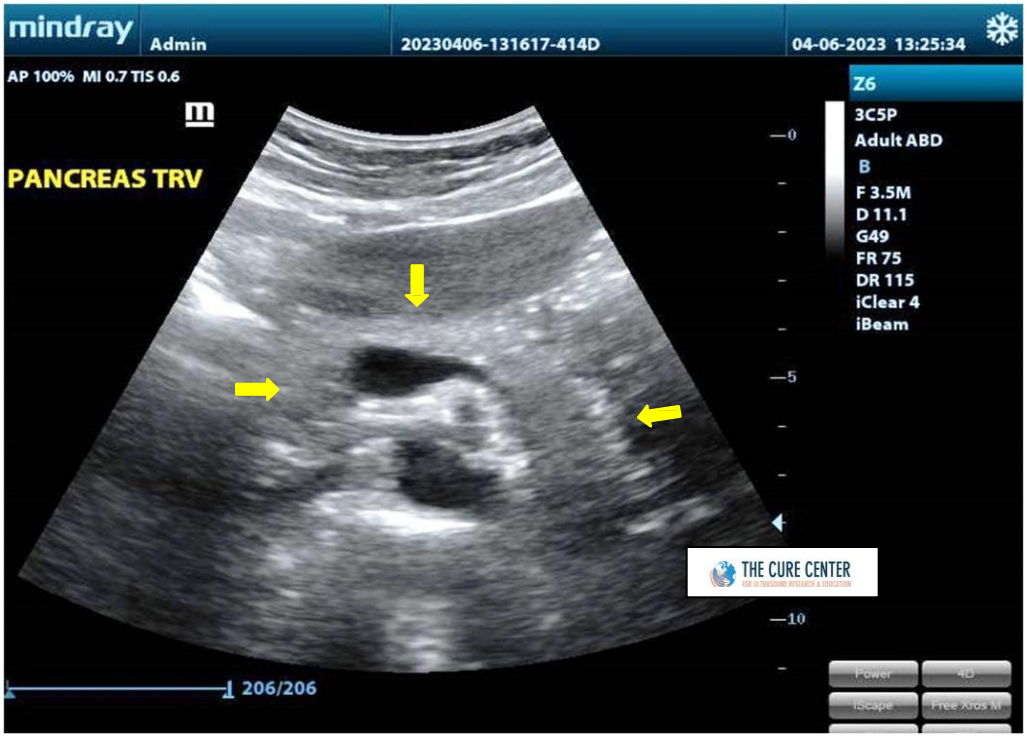


Pancreas Protocol

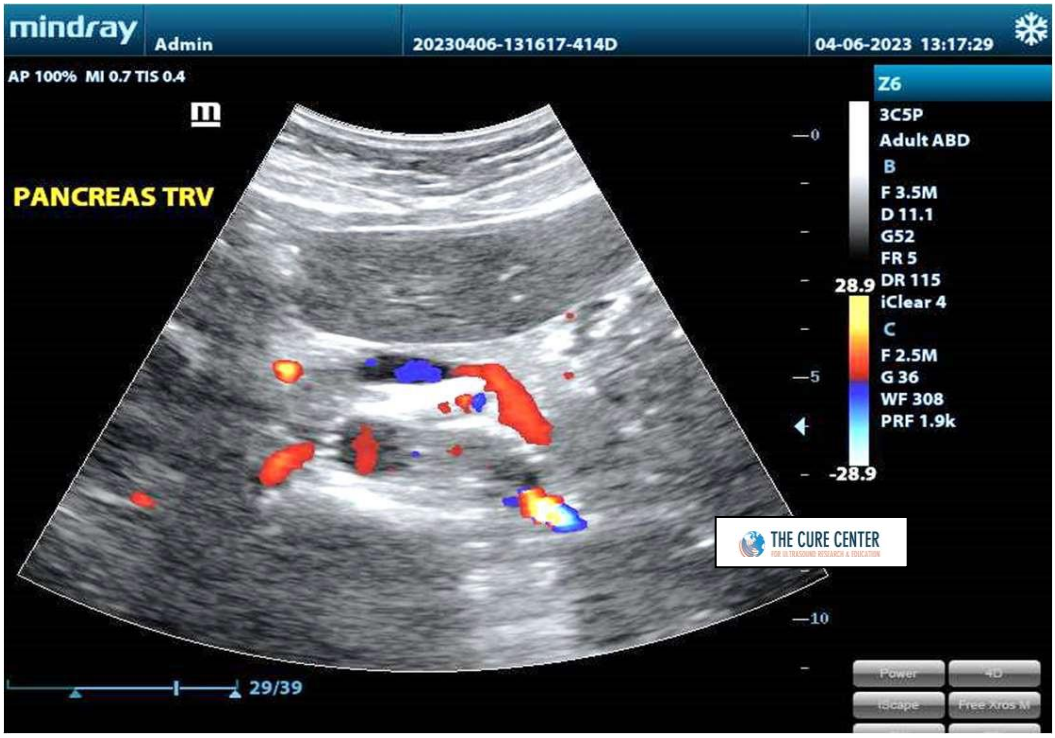
1. TRV Pancreas – Head/ Body/ Tail

Note:

The Transducer's notch should be toward you, with the other edge of the probe up toward the patient's left shoulder to get a full image of the pancreas.



2. Pancreas – Head/ Body/ Tail with Color Doppler



3. SAG Pancreas – Head

Note:

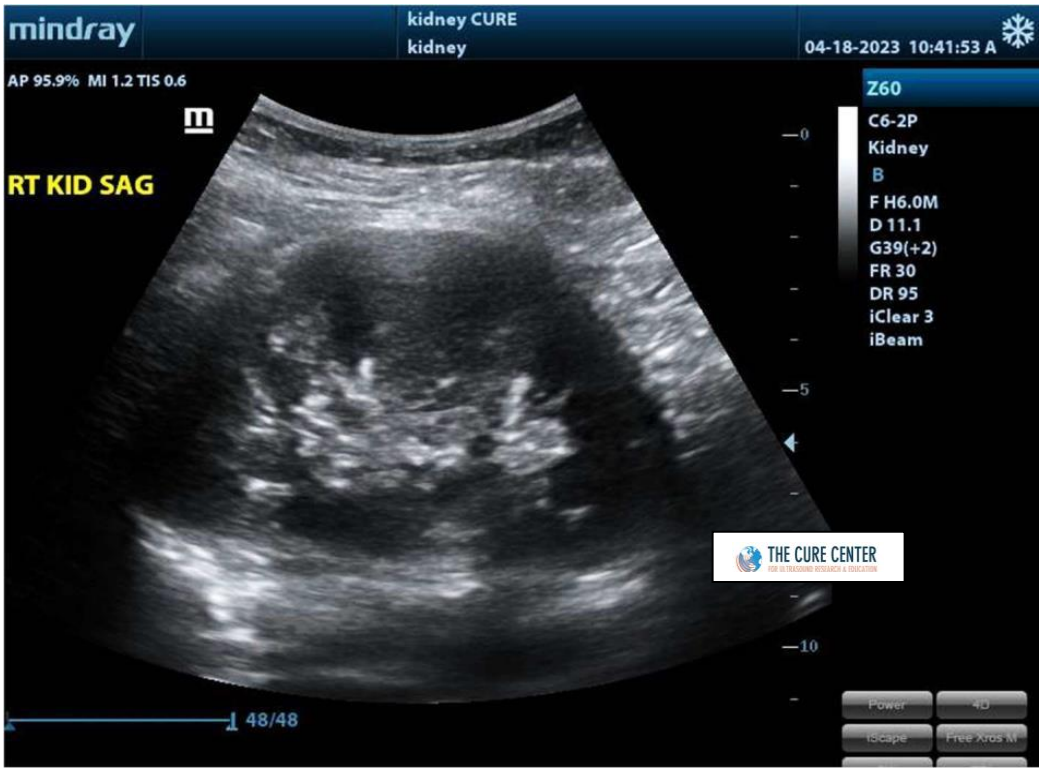
Make sure your PV is elongated; show the head of the pancreas above it. The head of the pancreas is between the liver and PV.



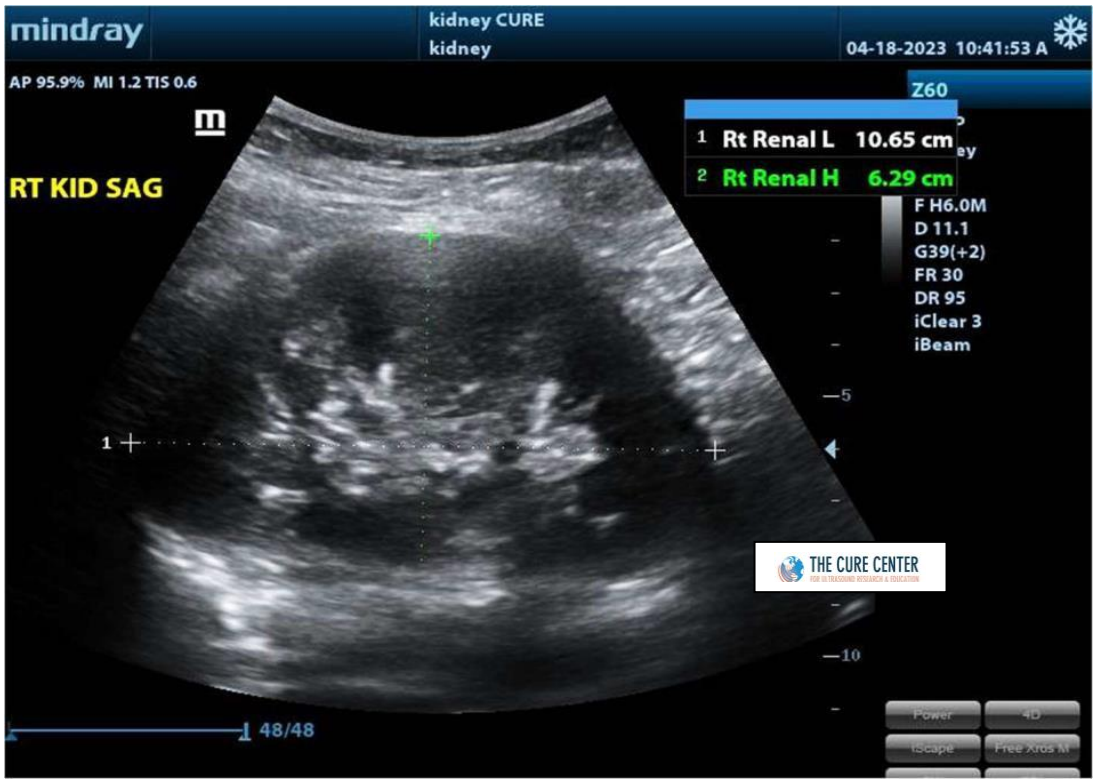
Kidney Protocol

Right Kidney Sagittal:

1. **RT KD SAG – Mid View Grayscale**



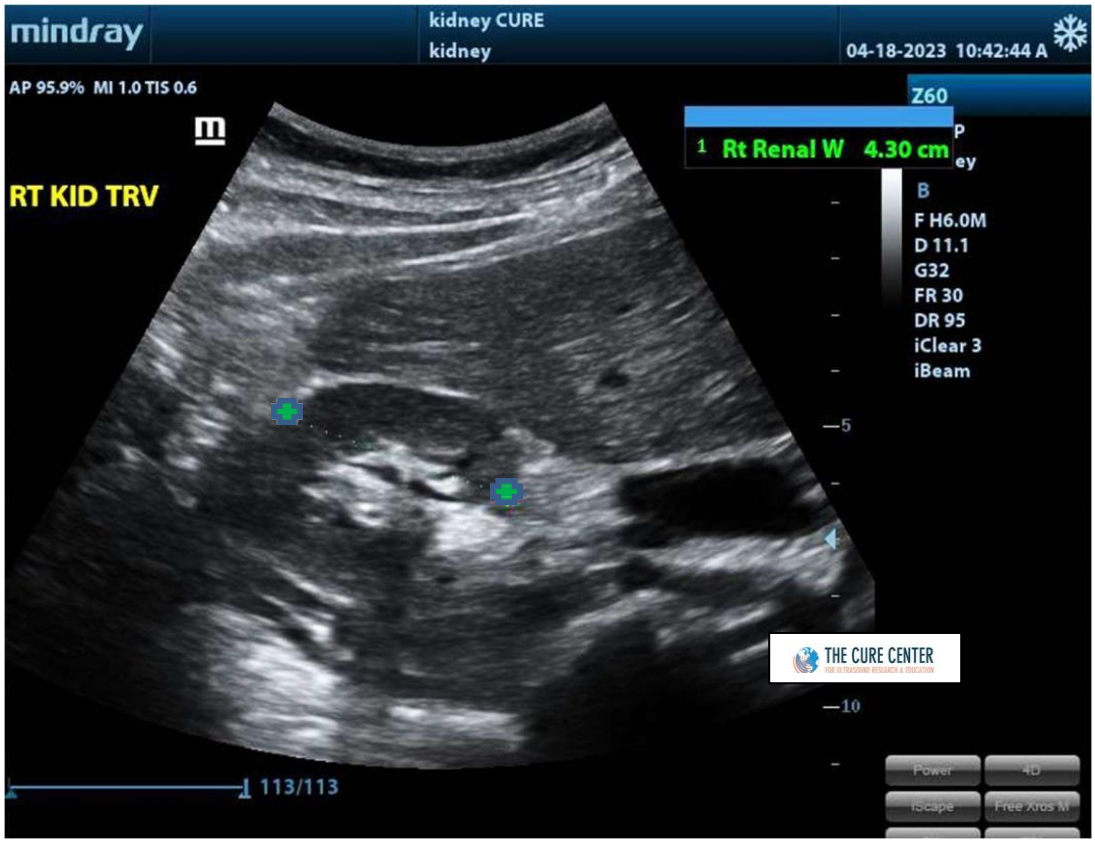
2. **RT KD SAG – Mid View w/ Measurements**



3. **RT KD TRV – Mid Pole**



4. **RT KD TRV – Mid Pole w/ Measurement**

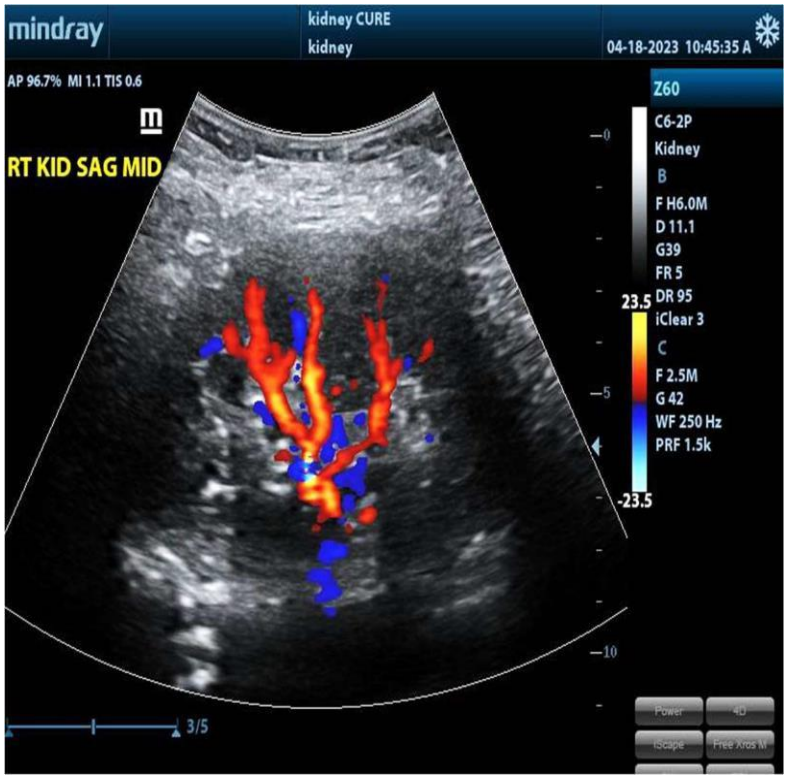


5. **RT KD SAG – Lateral Views**





6. **RT KD SAG- Mid View w/ Color Doppler**





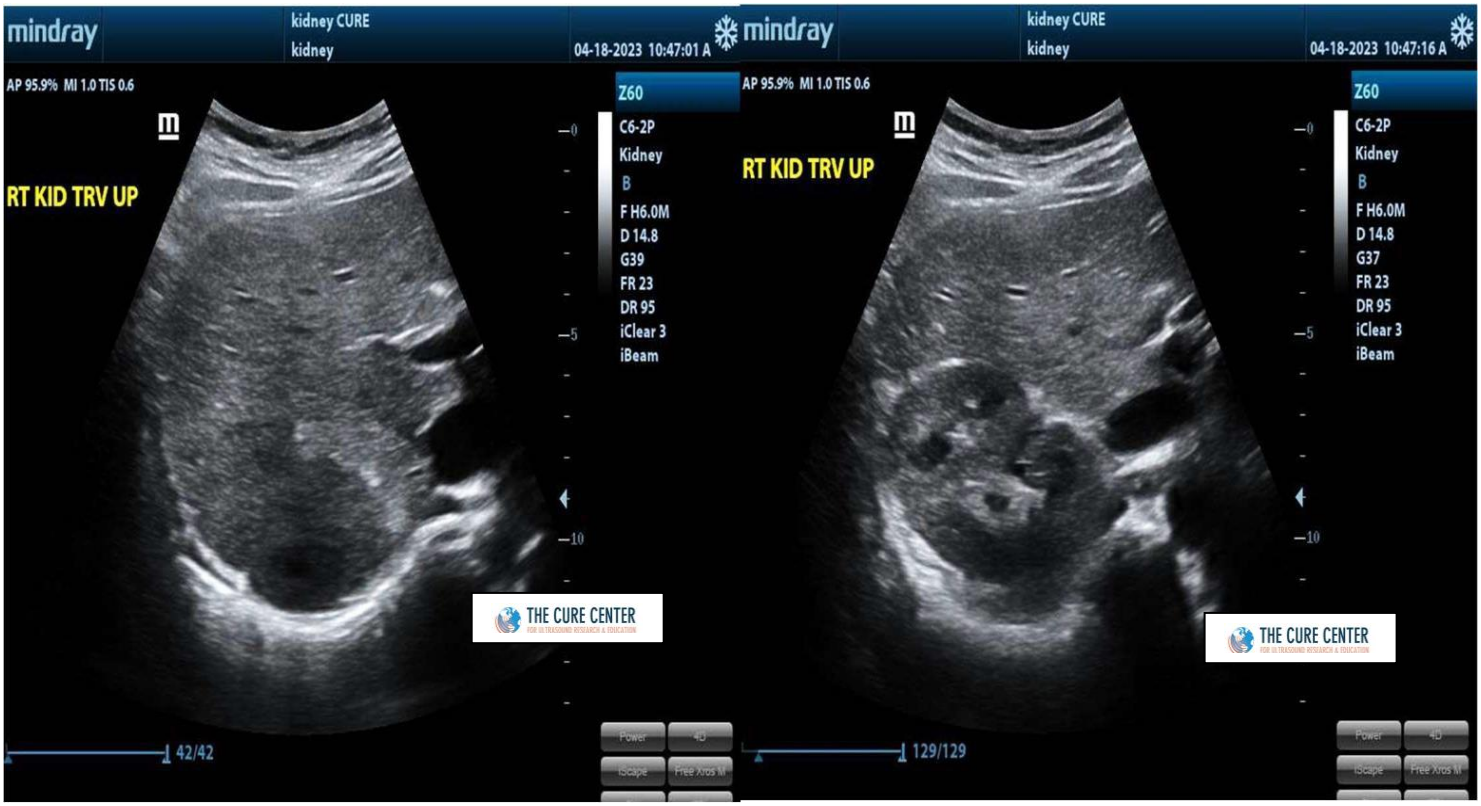
7. RT KD SAG – Medial View



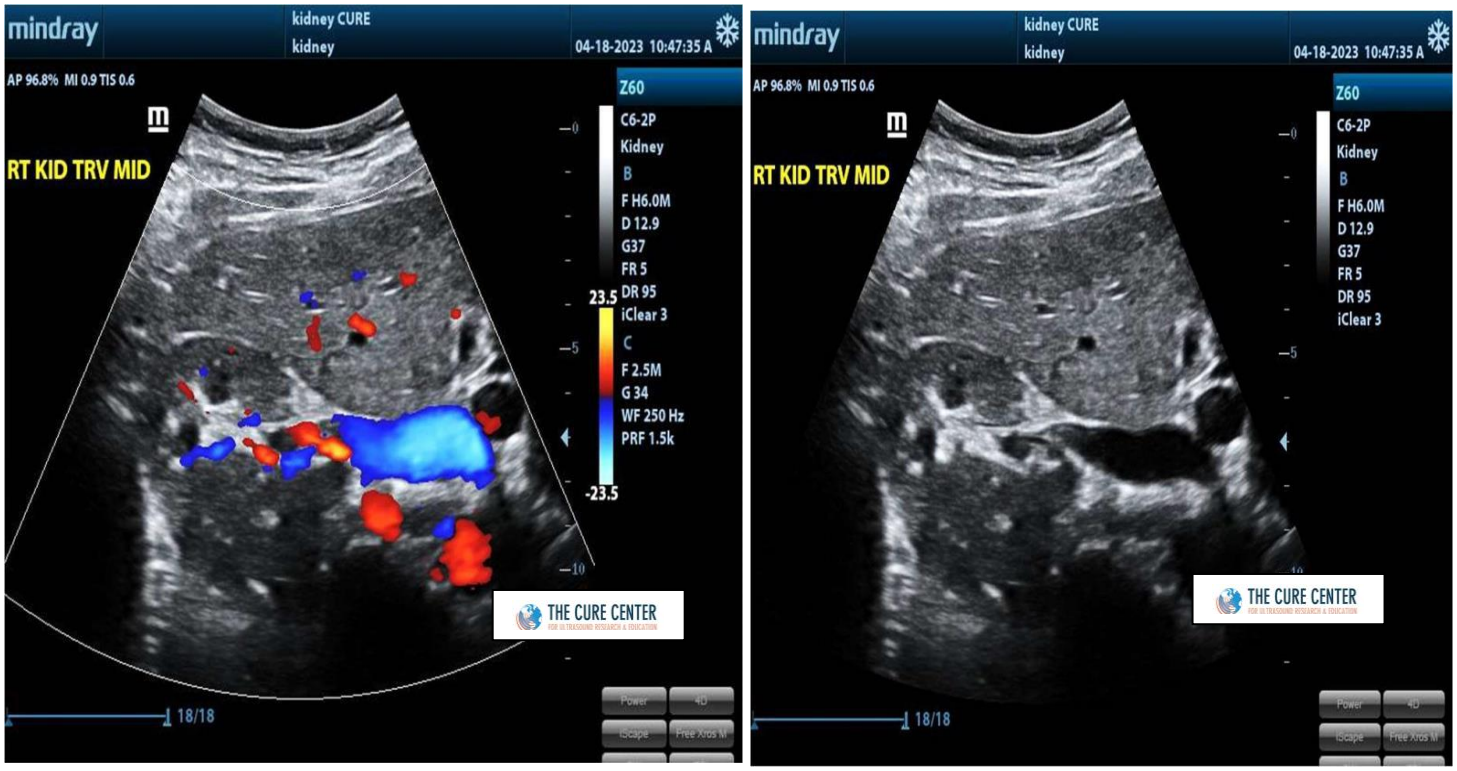


Right Kidney Transverse:

8- RT KD TRV – Upper Pole / Liver



9- RT KD TRV – Mid Pole w/ and w/o Color Doppler



10- RT KD TRV – Lower Pole



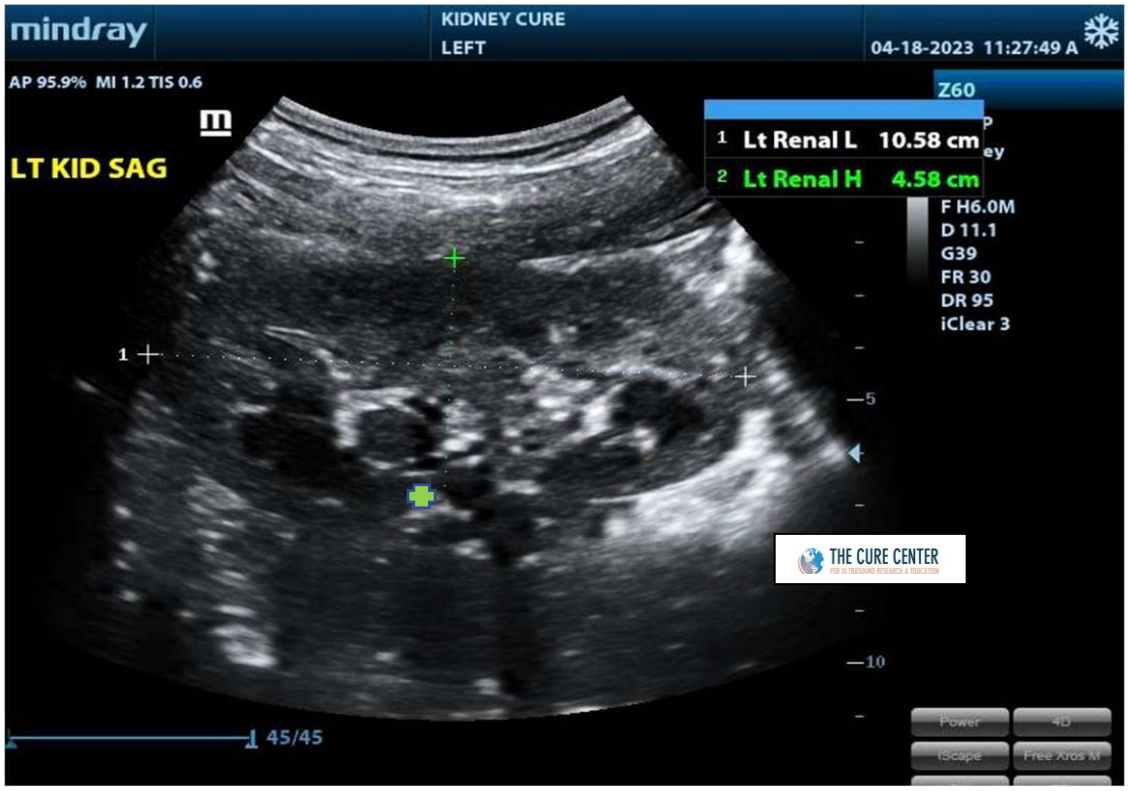


Left Kidney Sagittal:

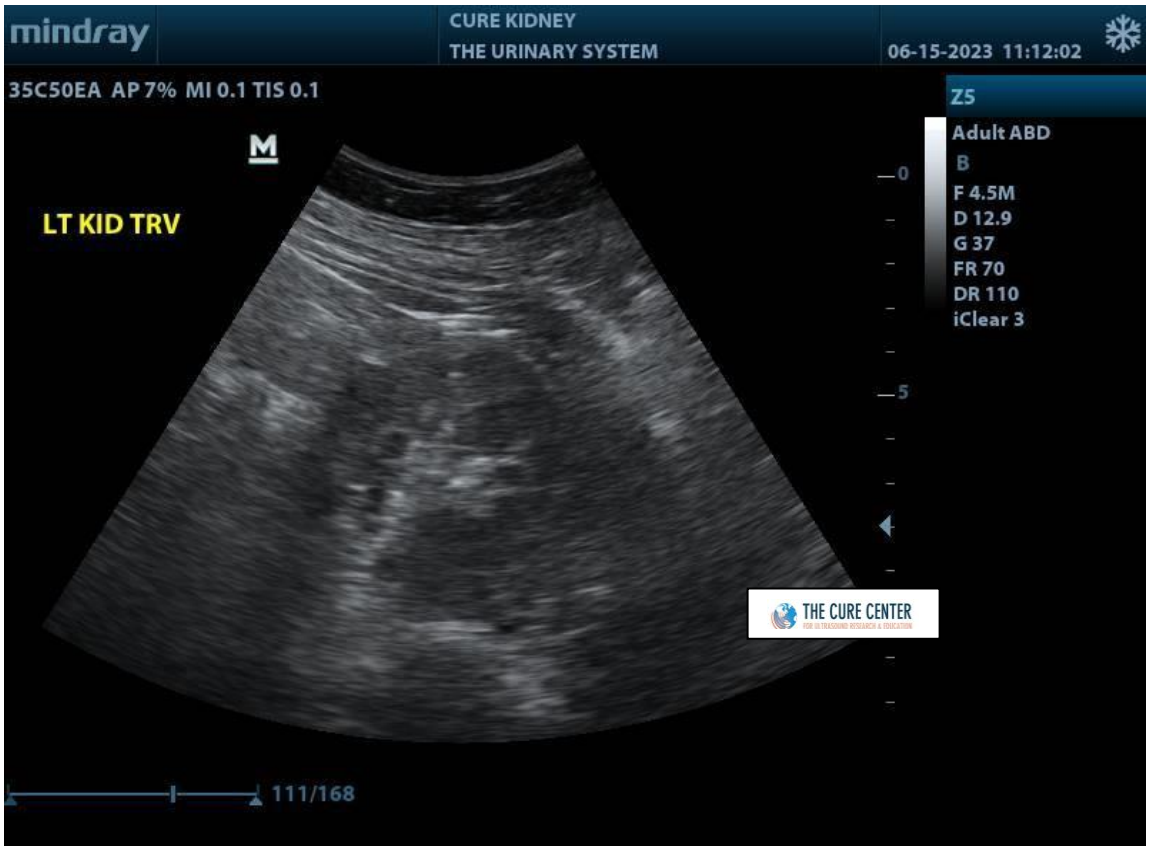
11- LT KD SAG – Mid View Grayscale



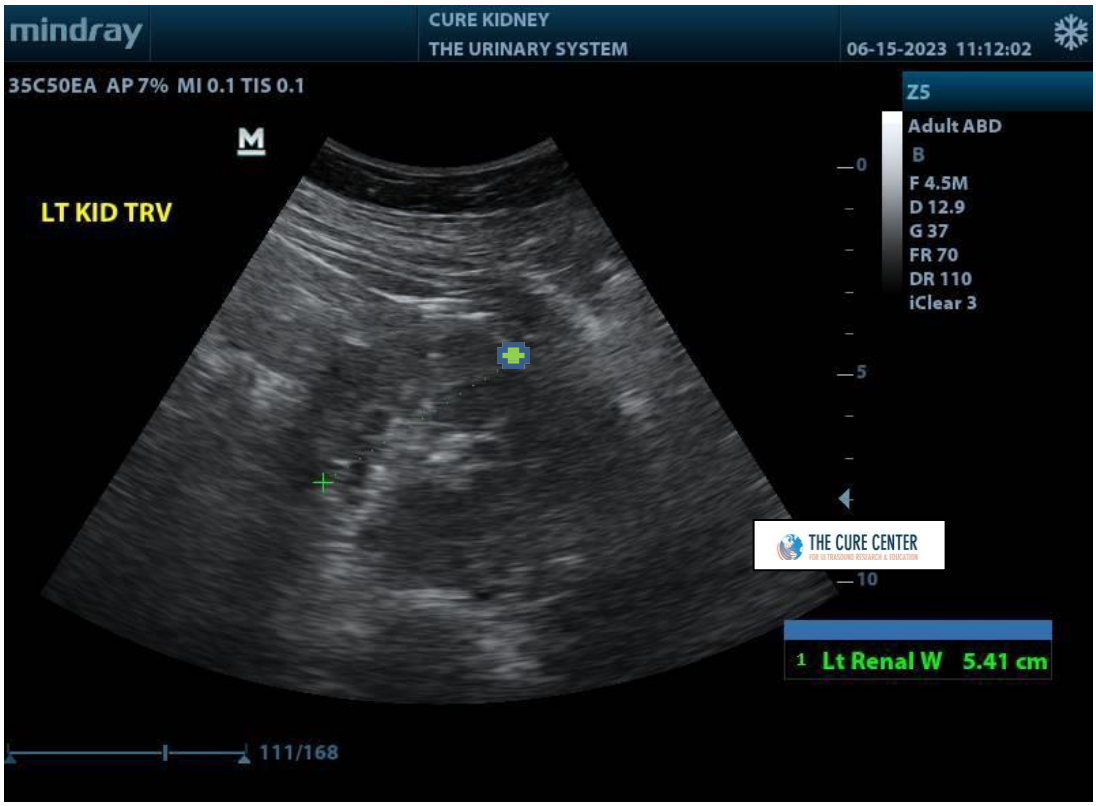
12- LT KD SAG – Mid View w/ Measurements



13- LT KD TRV– Mid View Grayscale



14- LT KD TRV – Mid View w/ Measurements

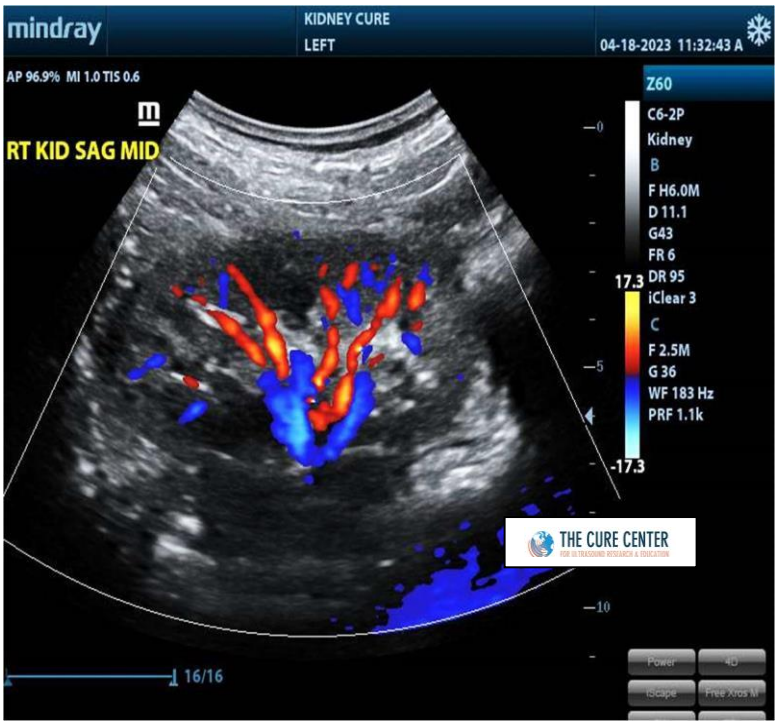


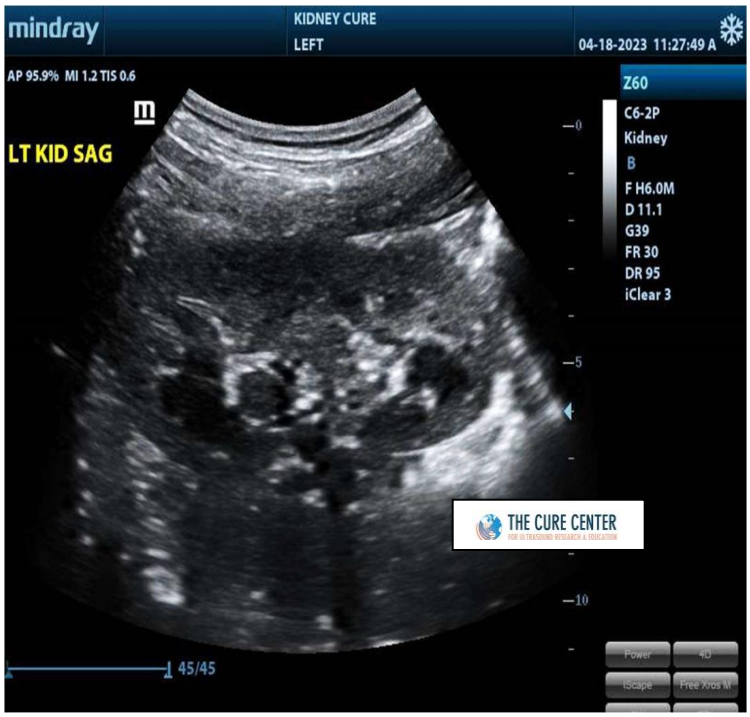
15- LT KD SAG – Lateral View





16- LT KD SAG – Mid View w/ and w/o Color Doppler





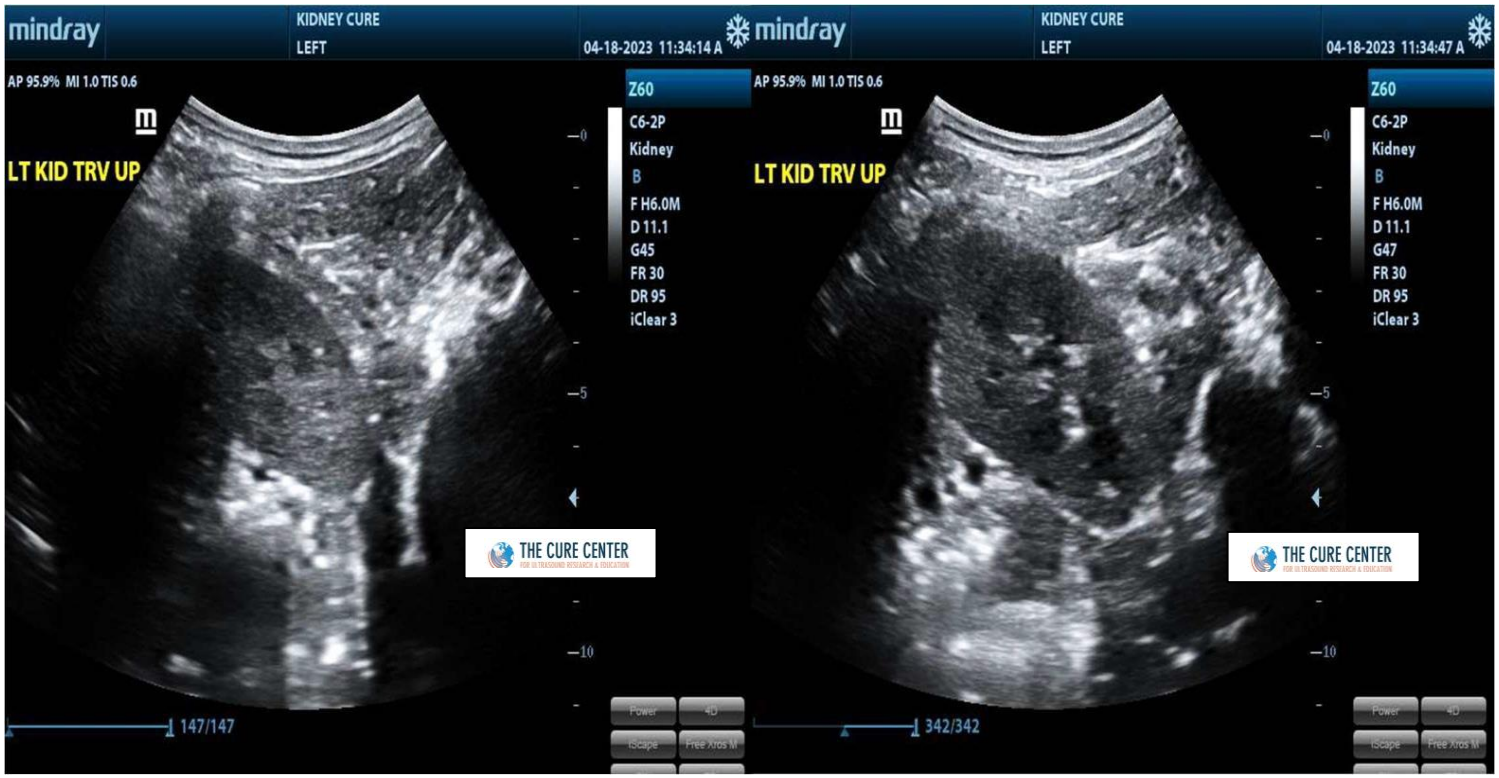
17- LT KD SAG -- Medial View



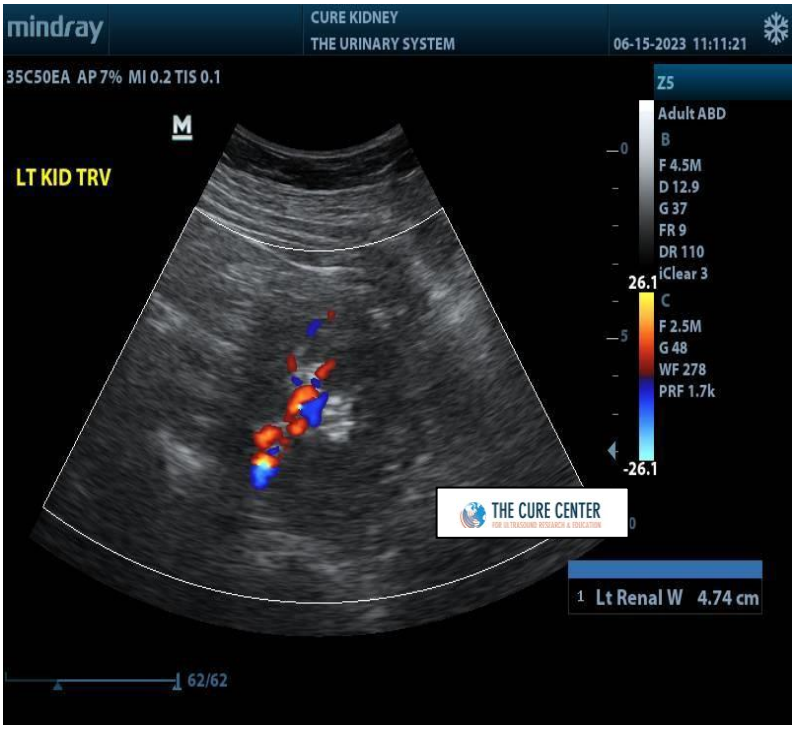


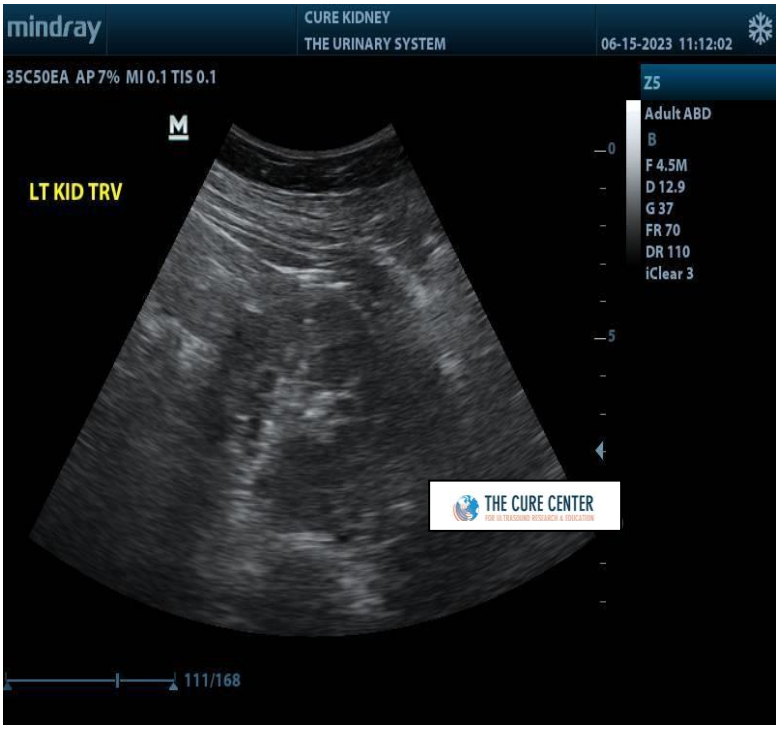
Left Kidney Transverse:

18- LT KD TRV – Upper Pole (some of the spleen might be seen in here)



19- LT KD TRV – Mid Pole w/ and w/o Color Doppler





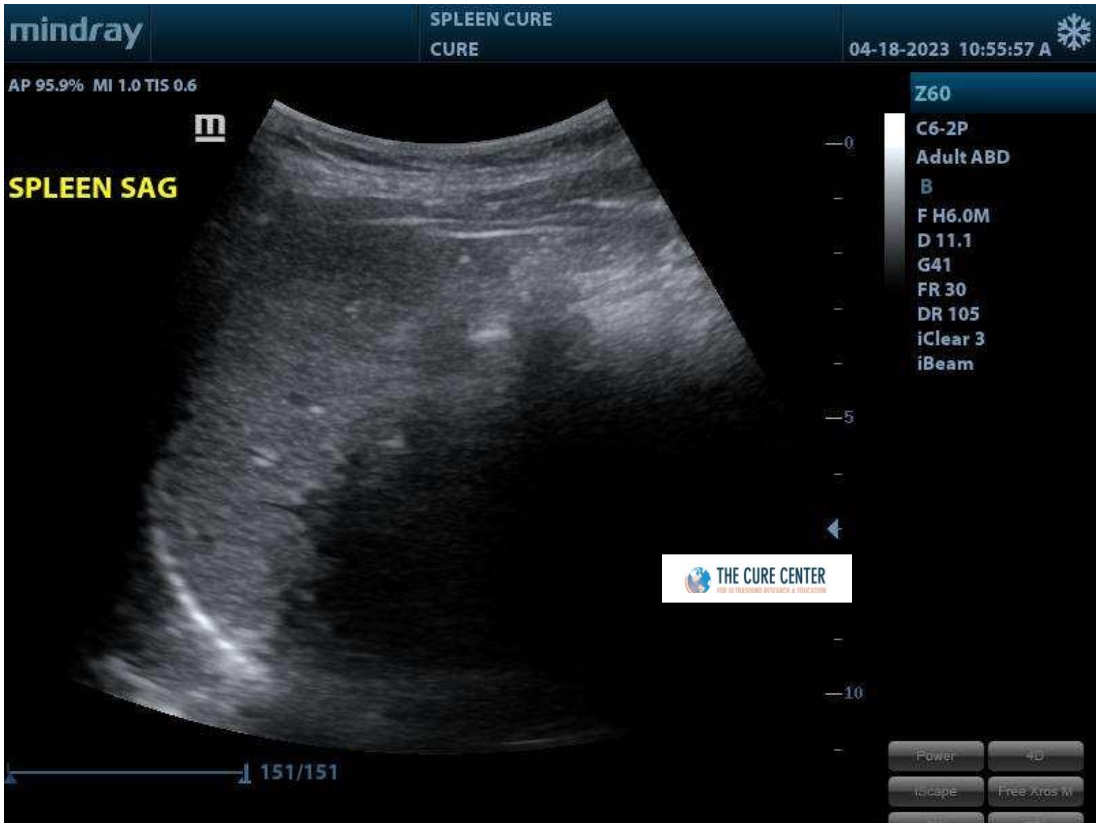
20- LT KD TRV – Lower Pole



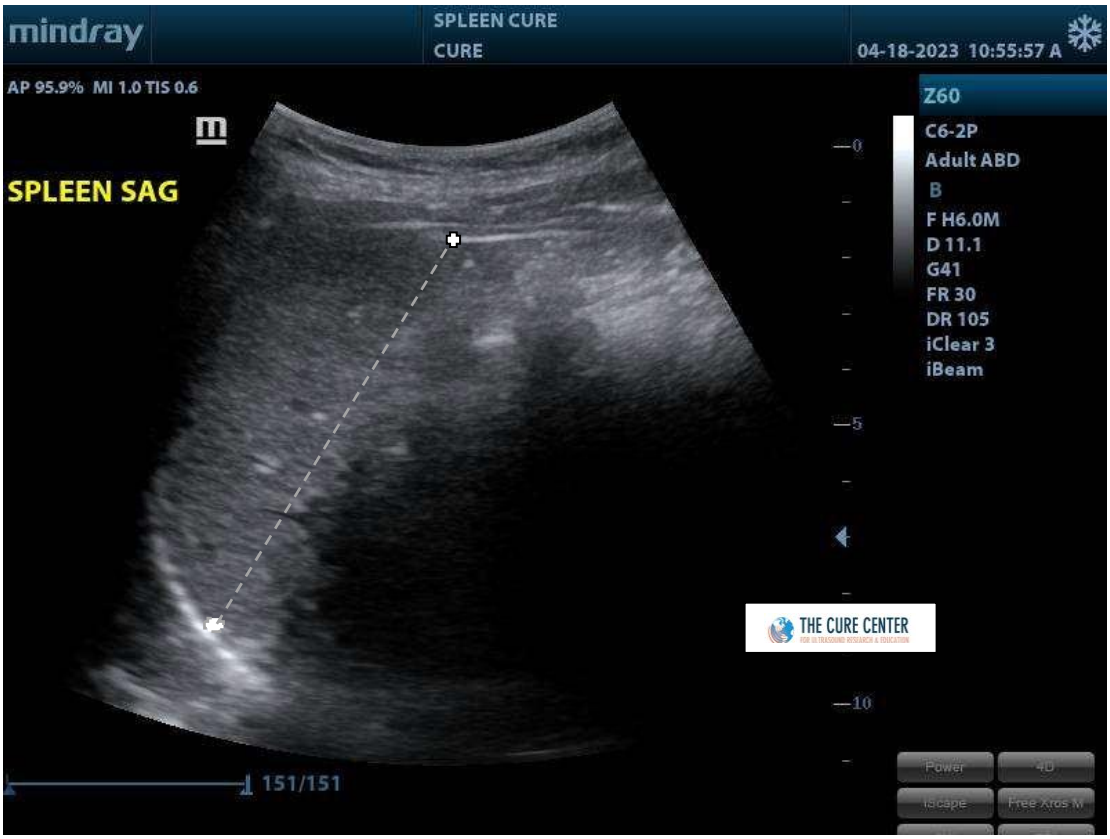


Spleen Protocol

1. SPL SAG – Grayscale



2. SPL SAG – Measurement 1x



3. SPL TRV – Grayscale



4. SPL TRV – Measurement 2x



5. SPL TRV – Color



Note: The echogenicity of the left kidney should be compared to the spleen echogenicity. If the spleen appears large, measure it in longitudinal and transverse planes. Image and measure any mass. If requested, and in all cirrhotic patients, evaluate the splenic vein and artery.